

TOTAL INSTRUMENT LOOPS	COLD LOOP INSPECT
560 Loops	100% Verifi
Master Client Schedule Rev 3.6a	Continuity, Megger 500VDC

1. LOOP TESTING SCOPE & SUBSYSTEM WORK BREAKDOWN

Subsystem / Plant Area	Target Field Enclosures	Associated PLC Rack
Jet Cooker Infeed & Pre-Treatment	JB-401 (Floor 2F)	Chassis C1 (Panel P1)
Jet Cooker Thermal Cooking Skid	JB-402 (Floor 2F)	Chassis C2 (Panel P2)
Spray Dryer Base & Powder Discharge	JB-601 (Floor 1F)	Chassis C1 (Panel P1)
Spray Dryer Cyclone & Exhaust Section	JB-602 (Floor 3F)	Chassis C2 (Panel P2)
Spray Dryer Direct Burner & Air Heater	JB-606 (Floor 6F)	Chassis C3 (Panel P3)
Spray Dryer Main Mid-Tower Section	JB-607, JB-608 (4F)	Chassis C2 & C3 (P2/P3)
Spray Dryer Upper Tower & Atomizer	JB-612 (Floor 6/8F)	Chassis C4 (Panel P4)
Packing Tower Feed & Silo Discharge	JB-618 (Packing)	Chassis C3 (Panel P3)
Hazardous Area Intrinsically Safe Loops	IS-JB-603/608/612/618	Chassis C4 (P4 IS)
Slurry Preparation Outbuilding	RIO-200 (Slurry 2F)	Chassis C5 (RIO-200)
Integrated Interlocks & Handover	CA1 Central MCP	Master Chassis C1-C5

2. REQUIRED TEST INSTRUMENTS, CALIBRATORS & COMMISSIONING TOOL SPECIFICATIONS

Item #	Equipment Description	Manufacturer & Model
1	Precision Documenting Process Calibrator	Fluke 754
2	Precision Digital Pressure Calibrator	Druck DPI 611
3	True RMS Digital Industrial Multimeter	Fluke 87V
4	High-Voltage Insulation Resistance Tester	Megger MIT410/2
5	Pneumatic Hand Test Pump w/ Vernier	Ralston Instruments AP-002
6	Industrial 2-Way VHF/UHF Radios	Motorola Mototrbo DP4801e

KALASIN ENGINEERING | xCIP-1545 AU

MASTER INSTRUMENT LOOP TESTING PLAN, PROCEDURE & COMMISSIONING

FUNCTION	HOT LOOP CALIBRATION	COMMISSIONING
Tested	5-Point Simulation	4 Weeks
Calibration, Polarity	0%, 25%, 50%, 75%, 100% Range	Mobilization

Total Loops	Discrete DI/DO	Analog AI/AO	Hazardous Area Zone
96	74	22	Safe Area
48	33	15	Safe Area
61	45	16	Safe Area
86	26	60	Safe Area
9	5	4	Classified / Burner
179	121	58	Dust Zone 22 (Ext)
40	32	8	Dust Zone 21/22
29	21	8	Dust Zone 22
99	50	49	ATEX Zone 1/21 (Ex ia)
66	48	18	Safe Area
560	334	226	Central Control

Calibration Parameter	Operating Range	Accuracy / Tolerance	Valid Calibration Certificate
4-20mA, Volts, RTD, TC, HART	-10 to 55 mA / 0-30V / Pt100	±0.01% of Reading	ISO/IEC 17025 Certified
Differential Pressure	-1 to 20 bar (Hydraulic/Pneu)	±0.0185% FS	ISO/IEC 17025 Certified
Resistance, Amps, Resistance, Conductivity	0 - 1000V DC / 0 - 20A / 50MΩ	±0.05% Accuracy	Factory Calibrated
Resistance @ 250V/500V/1kV	10 kΩ to 100 GΩ	±2% ±2 digits	Annual Certified
Pressure / Vacuum	0 to 300 psi (20 bar)	Fine Vernier 0.01 psi	Traceable Reference
Intrinsically Safe ATEX Comms	136 - 174 MHz	ATEX Gas/Dust Certified	Radio License Compliant

OTOMATION SYSTEM (Ref: x2608003)

MISSIONING SCHEDULE — SPRINT 18K TPA SPRAY DRYER / JET COOKER

MISSIONING SCHEDULE	STANDARD COMPLIANCE
Weeks (20 Days)	IEC 62381 / ISA
Commissioning to Client Handover	Field Instrument Loop Check Standard

Primary Test Protocol	Estimated Duration	Scheduled Window
Cold + 4-20mA Flow/Press Sim	3 Working Days	Week 1 (Day 3-5)
Temp RTD + Steam PRV Stroke	2 Working Days	Week 2 (Day 6-7)
Rotary Valve + Pressure Trans	2 Working Days	Week 2 (Day 8-9)
Diff Pressure DP + Temp RTDs	3 Working Days	Week 2 (Day 9-10)
Combustible Gas + Thermal Flow	1 Working Day	Week 3 (Day 11)
High-Density Transmitters & Valves	4 Working Days	Week 3 (Day 12-14)
Atomizer Speed + Vent Pressure	2 Working Days	Week 3 (Day 14-15)
Powder Silo Fork + Sieve DI	2 Working Days	Week 4 (Day 16-17)
P&F IS Barrier Insulation & Loop Check	3 Working Days	Week 4 (Day 17-18)
1Gbps DLR Ring + Slurry pH/Mass Flow	2 Working Days	Week 4 (Day 18-19)
Full ESD Trip & SCADA Graphic Verification	2 Working Days	Week 4 (Day 19-20)

Calibration Expiry Date	Dedicated Testing Purpose	Quantity Required
31-DEC-2026	Analog 4-20mA 5-Point Loop Simulation & HART Pressure & Differential Transmitter Calibration	2 Units
15-NOV-2026	Check continuity check, 24VDC loop power verification	1 Unit
28-FEB-2027	Pre-power cable insulation & shield isolation (>20MΩ)	3 Units
15-JAN-2027	Field Pressure Transmitter 5-point mechanical test	2 Units
31-DEC-2026	Direct Field-to-Control Room voice coordination	2 Units
Permanent		6 Units

PUNCHLIST TRACKING
Cat A & B System
<i>100% Pre-commissioning Sign-off</i>

Responsible Team Lead	Witness / QA/QC	Status
Lead Inst. Eng (Kalasin)	Ingredion QA/QC	READY FOR EXECUTION
Senior Comm. Eng	Ingredion QA/QC	READY FOR EXECUTION
Lead Inst. Eng (Kalasin)	Ingredion QA/QC	READY FOR EXECUTION
Senior Comm. Eng	Ingredion QA/QC	READY FOR EXECUTION
Lead Inst. Eng (Kalasin)	Burner Specialist	READY FOR EXECUTION
Lead Inst. Eng (Kalasin)	Ingredion QA/QC	READY FOR EXECUTION
Senior Comm. Eng	Ingredion QA/QC	READY FOR EXECUTION
Lead Inst. Eng (Kalasin)	Ingredion QA/QC	READY FOR EXECUTION
Ex-Certified Inspector	TUV / Client Safety	READY FOR EXECUTION
Senior Comm. Eng	Ingredion QA/QC	READY FOR EXECUTION
Project Commissioning Mgr	Ingredion Project Mgr	PLANNED

Serial Number	Certification Agency	Status
FLK-754-98 421	NIST / Thai Metrology	CERTIFIED ACTIVE
DRK-611-34 901	Baker Hughes Cal Lab	CERTIFIED ACTIVE
FLK-87V-11 204	Fluke Service Center	CERTIFIED ACTIVE
MEG-410-88 319	Megger Calibration Lab	CERTIFIED ACTIVE
RAL-AP-55 201	Ralston Metrology	CERTIFIED ACTIVE
MOT-DP-4801-SET	NBTC Thailand	INSPECTED ACTIVE

Stage #	Testing Activity / Protocol Step
STAGE 0	Pre-Testing Prerequisite & Safety LOTO
STAGE 1	Phase 1: Cold Loop Inspection & Cable Testing
STAGE 2	Phase 2: Hot Loop Check & Power-Up
STAGE 3	Phase 3: 5-Point Analog Loop Calibration Check (AI/AO)
STAGE 4	Phase 4: Discrete Digital Loop (DI) & Limit Switch Check
STAGE 5	Phase 5: Digital Output (DO) & Solenoid Valve Actuation
STAGE 6	Phase 6: Control Valve (AO/AI) Modulation & Positioner Stroke
STAGE 7	Phase 7: Alarm, Trip & Safety Interlock Verification
STAGE 8	Phase 8: Punchlist Management & Final Acceptance Handover

Detailed Standard Operating Procedure (SOP)

mechanical terminations are complete.

2. Obtain valid Work Permit (Hot/Cold Work) and apply Lockout/Tagout (LOTO) on pneumatic headers and high-voltage power feeds.

3. Verify calibration validity of Fluke 754 calibrators and Megger instruments.

4. Establish dedicated radio communication between Field Lead and

2. Verify cable core identification against marshaling wiring drawings (C1S4-19/P1-TBDI1-17).

3. Disconnect transmitter and I/O card terminals before insulation testing to prevent card damage.

4. Perform Megger test @ 500VDC: Core-to-Core and Core-to-Shield/Ground for minimum 60 seconds.

1. Reconnect field wiring to marshaling terminals and transmitter terminals.

2. Energize 24VDC power supply circuit breaker in Control Cabinet CA1.

3. Measure DC voltage across instrument terminals using Fluke 87V.

4. Measure quiescent current draw and confirm proper polarity (positive to + / return to -).

5. In intrinsically safe (IS) loops, verify Pepperl+Fuchs IS barrier output voltage and barrier grounding.

2. Inject current ascending: 0% (4.00 mA), 25% (8.00 mA), 50% (12.00 mA), 75% (16.00 mA), 100% (20.00 mA).

3. Record Raw PLC counts (0..32767 for 1756-IF16) and SCADA HMI graphical display reading in engineering units.

4. Repeat descending check: 100% -> 75% -> 50% -> 25% -> 0% to evaluate hysteresis.

2. For level forks (FTM51 / FTL51B): simulate tuning fork immersion / vibration damping.

3. For pressure switches (PTC31B) and flow switches (FSL): simulate trip pressure / paddle movement.

4. Observe LED status indicator on 1756-IB32 card and verify tag status change (0 -> 1) on SCADA screen.

2. Issue Manual DO command (Force or HMI manual toggle) from SCADA workstation.

3. Verify 1756-OB32 channel output energizes 24VDC interposing relay / solenoid valve coil.

4. Confirm physical actuator stroke movement and measure open/close stroke travel time with stopwatch.

Trovis 3730-1 positioner.

2. Command AO from SCADA at 0%, 25%, 50%, 75%, 100% and descending.

3. Measure physical valve travel scale indicator and compare against 4-20mA position transmitter feedback.

4. Test fail-safe action: simulate power loss and air supply disconnection; confirm valve drives to fail position (Fail-Closed FC or Fail-Open FO). (H/L, Low-Low (LL), High-High (HH)).

2. Confirm audible horn annunciator, alarm banner flashing, and historical alarm log timestamp.

3. Verify automated safety shutdown interlocks (e.g. Cyclone High Temp -> Trip Burner Gas Solenoid).

4. Confirm that latched emergency shutdown (ESD) alarms require manual operator reset.

2. Record punchlist item with clear tag ID, responsible contractor, and target completion date.

3. Affix tamper-evident waterproof green 'LOOP CHECK PASSED' calibration tag onto field instrument.

4. Compile signed loop test sheets, calibration records, and ITP documentation into Master Handover Binder.

RATING PROCEDURE (SOP) — INSTRUMENT LOOP TESTING

with IEC 62381 (Factory & Site Acceptance Testing of Automation Systems)

Pass / Acceptance Criteria	Responsible Personnel
Approved Work Permit, zero residual hazardous energy, test calibrator valid certificates on file.	Commissioning Lead & Safety Officer
Insulation resistance > 20 MΩ @ 500VDC. Shield grounded at Cabinet side ONLY (no double-grounding). Continuity resistance < 2 Ω.	Instrument Technician & QC Inspector
Terminal voltage within 21.6V to 26.4VDC. Quiescent current: 3.8 to 4.2 mA for analog 2-wire. No tripped fuses or circuit breaker faults.	Senior Commissioning Engineer
Max error ≤ ±0.5% of calibrated span across all 5 test points. Zero and span linearity verified. SCADA matches field simulation exactly.	Lead Instrument Engineer & SCADA Engineer
100% state change fidelity (Open=1, Close=0 or fail-safe logic). Card LED and SCADA display update within <100ms. No contact chatter.	Instrument Technician & SCADA Operator
Full valve travel stroke without sticking or pneumatic leakage. Stroke time within vendor datasheet specs (<3 sec for ball valves). SCADA graphic updates from Red to Green.	Commissioning Lead & Mechanical Tech
Travel error ≤ ±1.0% of full stroke. Hysteresis ≤ 0.5%. Fail-safe spring return drives to fail-safe position in <2.0 seconds.	Valve Specialist & Lead Inst Engineer
100% interlock trip action verified. ESD valve de-energization confirmed. Correct alarm priority colors (Red=Emergency, Yellow=Warning).	Lead Process Automation Engineer & Client Rep
Zero Category A punchlist items remaining. All Category B items agreed with sign-off. Green Loop Pass tag affixed to instrument.	Project Commissioning Manager & Client PM

Verifying Document / Record	QA/QC Hold Point
Work Permit + Equipment Cal Certs	HOLD POINT (Mandatory Sign-off)
Cold Loop Check Sheet (Sheet 03)	WITNESS POINT
Hot Loop Power-Up Record	REVIEW POINT
5-Point Calibration Record (Sheet 03)	HOLD POINT (Client Witness)
Discrete Test Log Sheet	WITNESS POINT
Valve Actuation Test Protocol	WITNESS POINT
Control Valve Calibration Certificate	HOLD POINT (Client Witness)
Safety Interlock Matrix Sign-off	HOLD POINT (Mandatory Sign-off)
Master Handover Certificate	FINAL ACCEPTANCE MILESTONE

TASK IDENTIFICATION & WORK BREAKDOWN			
Task ID	Activity / Loop Testing Scope Description	Target Area / JB	Duration
T-01	Mobilization, Safety Induction, LOTO & Calibrator Certification	Site-wide	2 Days
T-02	Control Cabinet CA1 Power-up & DLR Network Health Check	CA1 MCP	1 Day
T-03	JB-401 Jet Cooker Infeed Cold Loop Check & Megger Testing	JB-401 (2F)	2 Days
T-04	JB-401 Jet Cooker Hot Loop 5-Pt Calibration & Discrete Check	JB-401 (2F)	2 Days
T-05	JB-402 Jet Cooker Skid Cold Loop & Hot Loop Testing	JB-402 (2F)	2 Days
T-06	Jet Cooker Steam PRV & Samson Control Valve Stroke Check	Jet Cooker	1 Day
T-07	MILESTONE 1: Jet Cooker Skid Loop Test Handover Sign-off	Skid 2F	Milestone
T-08	JB-601 Spray Dryer Base Floor 1F Cold & Hot Loop Tests	JB-601 (1F)	2 Days
T-09	JB-602 Cyclone & Exhaust Floor 3F Cold & Hot Loop Tests	JB-602 (3F)	2 Days
T-10	JB-606 Direct Air Heater Burner Loop Checking	JB-606 (6F)	1 Day
T-11	JB-607 Mid-Tower Main Deck High-Density Loop Testing (Part 1)	JB-607 (4F)	2 Days
T-12	JB-607 Mid-Tower Main Deck High-Density Loop Testing (Part 2)	JB-607 (4F)	2 Days
T-13	JB-608 Auxiliary Deck Sight Glass & Level Testing	JB-608 (4F)	1 Day
T-14	JB-612 Upper Tower & Atomizer 6F/8F Loop Check	JB-612 (6/8F)	2 Days
T-15	MILESTONE 2: Spray Dryer Tower Loop Test Handover Sign-off	Dryer Tower	Milestone
T-16	JB-618 Packing Tower & Silos Cold & Hot Loop Tests	JB-618	2 Days
T-17	IS-JB-603 & IS-JB-608 Intrinsically Safe Ex ia Loop Check	IS JBs (Ex)	2 Days
T-18	IS-JB-612 & IS-JB-618 Intrinsically Safe Ex ia Loop Check	IS JBs (Ex)	2 Days
T-19	RIO-200 Slurry Outbuilding Remote Station Loop Check	RIO-200 (2F)	2 Days
T-20	Integrated System Safety Interlocks & ESD Trip Simulation	CA1 / Plant	2 Days
T-21	Category A & B Punchlist Clearance & Pre-start Rectification	Site-wide	1 Day
T-22	FINAL MILESTONE: Master Loop Check Protocol Tripartite Handover	Client MCP	Final

MASTER COMMISSIONING & LOOP TESTING EXECUTION SCHEDULE

Day (Days 1 to 20) Gantt Timeline Across All Plant Subsystems & Field Junction Boxes

Day (Days 1 to 20) Gantt Timeline Across All Plant Subsystems & Field Junction Boxes

[illegible]

Item #	Active Tag No.	Legacy Tag	P&ID No.
1	AT-60241	-	325-01-602-PD-05
2	DPT-60201	DPT-60901	325-01-602-PD-09
3	DPT-61301	-	325-01-613-PD-01
4	DPT-61302	-	325-01-613-PD-01
5	DPT-61310	-	325-01-613-PD-03
6	DPT-61312	-	325-01-613-PD-05
7	DPT-60202	-	325-01-602-PD-08
8	DPT-60203	-	325-01-602-PD-03
9	FSL-20201		325-01-202-PD-01
10	FSL-40201		325-01-402-PD-01
11	FSL-40202		325-01-402-PD-01
12	FS-40204		325-01-402-PD-01
13	FS-40205		325-01-402-PD-02
14	FT-40201	-	325-01-402-PD-02
15	FT-40202	-	325-01-402-PD-02
16	FT-40221		325-01-402-PD-02
17	FT-40222		325-01-402-PD-02
18	FT-60221	-	325-01-602-PD-04
19	FT-60201	-	325-01-602-PD-08
20	FT-60210	-	325-01-602-PD-03
21	FT-60211	-	325-01-602-PD-03
22	FT-60261FA	-	325-01-602-PD-02
23	FT-60261FB	-	325-01-602-PD-02
24	FT-60261FC	-	325-01-602-PD-02
25	FT-60261FD	-	325-01-602-PD-02
26	FT-60262FE	-	325-01-602-PD-02
27	FT-60262FF	-	325-01-602-PD-02
28	FT-60262FG	-	325-01-602-PD-02
29	FT-60262FH	-	325-01-602-PD-02
30	FT-60263FJ	-	325-01-602-PD-02
31	FT-60263FK	-	325-01-602-PD-02

32	FT-60263FL	-	325-01-602-PD-02
33	FT-60263FM	-	325-01-602-PD-02
34	FT-60264FN	-	325-01-602-PD-02
35	FT-60264FP	-	325-01-602-PD-02
36	FT-60264FQ	-	325-01-602-PD-02
37	FT-60264FR	-	325-01-602-PD-02
38	FT-61307	-	325-01-613-PD-04
39	FT-60203	FT-60901	325-01-602-PD-09
40	FT-60202	FT-60902	325-01-602-PD-09
41	FT-60204	FT-61101	325-01-602-PD-10
42	FT-936201	FT-90301	325-01-930-PD-01
43	FT-936202	FT-93001	325-01-930-PD-02
44	HS-61307A	-	325-01-613-PD-03
45	HS-61307B	-	325-01-613-PD-03
46	HS-61307C	-	325-01-613-PD-03
47	HS-61307D	-	325-01-613-PD-03
48	HS-61307E	-	325-01-613-PD-03
49	HS-61307F	-	325-01-613-PD-03
50	HS-61307G	-	325-01-613-PD-03
51	HS-61307H	-	325-01-613-PD-03
52	HS-61307I	-	325-01-613-PD-03
53	HS-61307J	-	325-01-613-PD-03
54	HS-61307K	-	325-01-613-PD-03
55	HS-61307L	-	325-01-613-PD-03
56	HS-61307M	-	325-01-613-PD-03
57	HS-61307N	-	325-01-613-PD-03
58	HS-61307O	-	325-01-613-PD-03
59	HS-61307P	-	325-01-613-PD-03
60	HS-61307Q	-	325-01-613-PD-03
61	HS-61307R	-	325-01-613-PD-03
62	HV-40203	HV-4106	325-01-402-PD-01
63	HV-40201	HV-4108	325-01-402-PD-01
64	HV-40204	HV-4126	325-01-402-PD-01
65	HV-40202	HV-4128	325-01-402-PD-01
66	HV-40206	HV-4208	325-01-402-PD-01

67	HV-40205	HV-4210	325-01-402-PD-01
68	HV-40207		325-01-402-PD-01
69	HV-40208		325-01-402-PD-01
70	LSH-60230	-	325-01-602-PD-04
71	LSH-60231	-	325-01-602-PD-04
72	LSH-60240	-	325-01-602-PD-05
73	LSH-60250	-	325-01-602-PD-05
74	LSH-60251	-	325-01-602-PD-04
75	LSL-60201	LSL-61101	325-01-602-PD-11
76	LSH-61301	-	325-01-613-PD-01
77	LSL-61302	-	325-01-613-PD-01
78	LS-61303	-	325-01-613-PD-01
79	LSH-61304	-	325-01-613-PD-03
80	LSH-61305	-	325-01-613-PD-05
81	LSH-61306	-	325-01-613-PD-01
82	LSH-61307		325-01-613-PD-01
83	LSH-61308		325-01-613-PD-01
84	LSH-61312	-	325-01-613-PD-05
85	LSL-814201	LSL-81001	325-01-814-PD-01
86	LT-40201	LT-40101	325-01-402-PD-01
87	LT-40202	LT-40102	325-01-402-PD-01
88	LT-40203	LT-40201	325-01-402-PD-02
89	LT-61301	-	325-01-613-PD-01
90	pHT-40201	pHT-40101	325-01-402-PD-01
91	pHT-40202	pHT-40102	325-01-402-PD-01
92	PSL-60220	-	325-01-602-PD-04
93	PSL-61301		325-01-613-PD-01
94	PSL-61302	-	325-01-613-PD-01
95	PSL-61305	-	325-01-613-PD-03
96	PSL-61306	-	325-01-613-PD-05
97	PSL-61311	-	325-01-613-PD-04
98	PT-40201	-	325-01-402-PD-02
99	PT-40202	-	325-01-402-PD-02
100	PT-40222		325-01-402-PD-02
101	PT-60201	-	325-01-602-PD-02

102	PT-60202	-	325-01-602-PD-01
103	PT-60203	-	325-01-602-PD-08
104	PT-60210	-	325-01-602-PD-03
105	PT-60215	-	325-01-602-PD-03
106	PT-60216	-	325-01-602-PD-03
107	PT-60220	-	325-01-602-PD-04
108	PT-60221	-	325-01-602-PD-04
109	PT-60230	-	325-01-602-PD-04
110	PT-60231	-	325-01-602-PD-04
111	PT-60240	-	325-01-602-PD-05
112	PT-60241	-	325-01-602-PD-05
113	PT-60204	PT-61101	325-01-602-PD-10
114	PT-61301	-	325-01-613-PD-01
115	PT-61302	-	325-01-613-PD-03
116	PT-61303	-	325-01-613-PD-01
117	PT-61304	-	325-01-613-PD-01
118	PT-61305	-	325-01-613-PD-01
119	PT-61309	-	325-01-613-PD-04
120	PCV-60201	PV-60201	325-01-602-PD-02
121	SV-40201	SV-40101	325-01-402-PD-01
122	SV-40202	SV-40102	325-01-402-PD-01
123	SV-40203	SV-40103	325-01-402-PD-01
124	SV-40204	SV-40104	325-01-402-PD-01
125	SV-60220	-	325-01-602-PD-04
126	SV-60230	-	325-01-602-PD-04
127	SV-60231	-	325-01-602-PD-04
128	SV-60240A	-	325-01-602-PD-05
129	SV-60240B	-	325-01-602-PD-05
130	SV-60240C	-	325-01-602-PD-05
131	SV-61301	-	325-01-613-PD-01
132	SV-61302	-	325-01-613-PD-01
132	SV-61306		325-01-613-PD-01
133	SV-61303	-	325-01-613-PD-01
134	SV-61305	-	325-01-613-PD-01
135	SV-61308	-	325-01-613-PD-03

136	SV-61309	-	325-01-613-PD-03
137	SV-61310	-	325-01-613-PD-05
138	SV-61311	-	325-01-613-PD-05
139	TSH-60230	-	325-01-602-PD-04
140	TSH-60240	-	325-01-602-PD-05
141	TT-40221		325-01-402-PD-02
142	TT-40222		325-01-402-PD-02
143	TT-60201	-	325-01-602-PD-02
144	TT-60210	-	325-01-602-PD-03
145	TT-60210A	TT-60212A	325-01-602-PD-03
146	TT-60210B	TT-60212B	325-01-602-PD-03
147	TT-60211	-	325-01-602-PD-03
148	TT-60214	-	325-01-602-PD-03
149	TT-60215A	-	325-01-602-PD-03
150	TT-60215B	-	325-01-602-PD-03
151	TT-60216	-	325-01-602-PD-03
152	TT-60220	-	325-01-602-PD-04
153	TT-60221	-	325-01-602-PD-04
154	TT-60222	-	325-01-602-PD-04
155	TT-60223	-	325-01-602-PD-04
156	TT-60224	-	325-01-602-PD-04
157	TT-60225	-	325-01-602-PD-04
158	TT-60226	-	325-01-602-PD-04
159	TT-60227	-	325-01-602-PD-04
160	TT-60230	-	325-01-602-PD-04
161	TT-60240	-	325-01-602-PD-05
162	TT-60241A	-	325-01-602-PD-05
163	TT-60241B	-	325-01-602-PD-05
164	TT-60245	-	325-01-602-PD-05
165	TT-60202	TT-60702	325-01-602-PD-07
166	TT-61301	-	325-01-613-PD-01
167	TT-61302	-	325-01-613-PD-01
168	TCV-40222		325-01-402-PD-02
169	TCV-60214	TV-60214	325-01-602-PD-03
170	TCV-60202	TV-60702	325-01-602-PD-07

171	VT-60210A	-	325-01-602-PD-03
172	VT-60210B	-	325-01-602-PD-03
173	VT-60241A	-	325-01-602-PD-05
174	VT-60241B	-	325-01-602-PD-05
175	XT-60201	XT-60301	325-01-402-PD-03
176	XA-60201	XA-60301	325-01-402-PD-03
177	XT-92201	-	325-01-923-PD-03
178	XA-92201	-	325-01-923-PD-03
179	XV-40201A	XV-40101A	325-01-402-PD-01
180	XV-40201B	XV-40101B	325-01-402-PD-01
181	XV-40201C	XV-40101C	325-01-402-PD-01
182	XV-40201D	XV-40101D	325-01-402-PD-01
183	XV-40201E	XV-40101E	325-01-402-PD-01
184	XV-40201F	XV-40101F	325-01-402-PD-01
185	XV-40201G	XV-40101G	325-01-402-PD-01
186	XV-40201H	XV-40101H	325-01-402-PD-01
187	XV-40202A	XV-40102A	325-01-402-PD-01
188	XV-40202B	XV-40102B	325-01-402-PD-01
189	XV-40202C	XV-40102C	325-01-402-PD-01
190	XV-40202D	XV-40102D	325-01-402-PD-01
191	XV-40202E	XV-40102E	325-01-402-PD-01
192	XV-40202F	XV-40102F	325-01-402-PD-01
193	XV-40202G	XV-40102G	325-01-402-PD-01
194	XV-40202H	XV-40102H	325-01-402-PD-01
195	XV-40203B	XV-40201B	325-01-402-PD-02
196	XV-40203C	XV-40201C	325-01-402-PD-02
197	XV-40203D	XV-40201D	325-01-402-PD-02
198	XV-40203E	XV-40201E	325-01-402-PD-02
199	XV-40203F		325-01-402-PD-02
200	XV-40204B		325-01-402-PD-01
201	XV-40204C		325-01-402-PD-01
202	XV-40205A		325-01-402-PD-02
203	XV-40205B		325-01-402-PD-02
204	XV-40205C		325-01-402-PD-02
205	XV-40205D		325-01-402-PD-02

206	XV-40205F		325-01-402-PD-02
207	XV-40222		325-01-402-PD-02
208	XV-60201	-	325-01-602-PD-02
209	XV-60251	-	325-01-602-PD-01
210	XV-60214	XV-60301	325-01-602-PD-03
211	XV-60202	XV-61101	325-01-602-PD-11
212	XV-61301	HV-61301	325-01-613-PD-01
213	XV-61302	-	325-01-613-PD-02
214	XV-61303	-	325-01-613-PD-02
215	DV-61301		325-01-613-PD-01
216	DV-61302		325-01-613-PD-01
217	ZS-60201A	-	325-01-602-PD-04
218	ZS-60201B	-	325-01-602-PD-04
219	ZS-60201C	-	325-01-602-PD-04
220	ZS-60201D	-	325-01-602-PD-04
221	ZS-60201E	-	325-01-602-PD-04
222	ZS-60229		325-01-602-PD-04
223	ZS-60230		325-01-602-PD-04
224	ZS-60231A	-	325-01-602-PD-05
225	ZS-60231B	-	325-01-602-PD-05
226	ZS-60231C	-	325-01-602-PD-05
227	ZS-60231D	-	325-01-602-PD-05
228	ZS-60231E	-	325-01-602-PD-05
229	ZS-60231F	-	325-01-602-PD-05
230	ZS-60240		325-01-602-PD-05
231	ZS-61310	-	325-01-613-PD-01
232	ZS-61311	-	325-01-613-PD-01
233	ZS-61312	-	325-01-613-PD-01
234	ZS-61313	-	325-01-613-PD-01
235		-	-
236		-	-
237		-	-
238		-	-
239		-	-
240		-	-

241		-	-
242		-	-
243		-	-
244		-	-
245		-	-
246		-	-
246	ES-60201		-
246	ES-60202		-
246	ES-60203		-
246	ES-61301		-
247	DEDERT SUPPLY EQUIPMENT	-	-
248	FT-60207	FT-60701	325-01-602-PD-07
249	PT-60207	PT-60701	325-01-602-PD-07
250	XV-60240D	SV-60240D	325-01-602-PD-05
251	SV-60207	SV-60701	325-01-602-PD-07
252	SV-60201	-	325-01-602-PD-08
	SSL-60201		325-01-602-PD-08
	ZS-60201		325-01-602-PD-08
253	SV-60202	-	325-01-602-PD-08
	SSL-60202		325-01-602-PD-08
	ZS-60202		325-01-602-PD-08
254	SV-60203	-	325-01-602-PD-08
	SSL-60203		325-01-602-PD-08
	ZS-60203		325-01-602-PD-08
255	SV-61304	-	325-01-613-PD-01
	SSL-61304		325-01-613-PD-01
	ZS-61304		325-01-613-PD-01
256	TT-60207	TT-60701	325-01-602-PD-07
257	XV-60261FA	-	325-01-602-PD-02
258	XV-60261FB	-	325-01-602-PD-02
259	XV-60261FC	-	325-01-602-PD-02
260	XV-60261FD	-	325-01-602-PD-02
261	XV-60261SA	-	325-01-602-PD-02
262	XV-60261SB	-	325-01-602-PD-02
263	XV-60261SC	-	325-01-602-PD-02

264	XV-60261SD	-	325-01-602-PD-02
265	XV-60262FE	-	325-01-602-PD-02
266	XV-60262FF	-	325-01-602-PD-02
267	XV-60262FG	-	325-01-602-PD-02
268	XV-60262FH	-	325-01-602-PD-02
269	XV-60262SE	-	325-01-602-PD-02
270	XV-60262SF	-	325-01-602-PD-02
271	XV-60262SG	-	325-01-602-PD-02
272	XV-60262SH	-	325-01-602-PD-02
273	XV-60263FJ	-	325-01-602-PD-02
274	XV-60263FK	-	325-01-602-PD-02
275	XV-60263FL	-	325-01-602-PD-02
276	XV-60263FM	-	325-01-602-PD-02
277	XV-60263SJ	-	325-01-602-PD-02
278	XV-60263SK	-	325-01-602-PD-02
279	XV-60263SL	-	325-01-602-PD-02
280	XV-60263SM	-	325-01-602-PD-02
281	XV-60264FN	-	325-01-602-PD-02
282	XV-60264FP	-	325-01-602-PD-02
283	XV-60264FQ	-	325-01-602-PD-02
284	XV-60264FR	-	325-01-602-PD-02
285	XV-60264SN	-	325-01-602-PD-02
286	XV-60264SP	-	325-01-602-PD-02
287	XV-60264SQ	-	325-01-602-PD-02
288	XV-60264SR	-	325-01-602-PD-02
289	XV-60261DA		325-01-602-PD-02
290	XV-60262DE		325-01-602-PD-02
291	XV-60263DN		325-01-602-PD-02
292	XV-60264DJ		325-01-602-PD-02
293	XV-60241	-	325-01-602-PD-05
294	XV-60240	-	325-01-602-PD-08
295	Machine Equipment Supply	-	-
296	PSE-61300	-	325-01-613-PD-01
297	PSE-61301	-	325-01-613-PD-01
298	ISV-61302	-	325-01-613-PD-01

299	PSE-61302	-	325-01-613-PD-03
300	PSE-61303	-	325-01-613-PD-05
301	PSE-61304	-	325-01-613-PD-01
302	ZS-61306	PSV-61301A	325-01-613-PD-01
303	ZS-61307	PSV-61301B	325-01-613-PD-01
304	ISV-61303	-	325-01-613-PD-03
305	ISV-61304	-	325-01-613-PD-05
306	MCC ROOM	-	-
307	AT-40201	ATY-40101	325-01-402-PD-01
308	AT-40202	ATY-40102	325-01-402-PD-01
309	AT-40203	ATY-40201	325-01-402-PD-02
310	BF-60210	BFY-60210	325-01-602-PD-03
311	BF-60211	BFY-60211	325-01-602-PD-03
312	BF-60240	BFY-60240	325-01-602-PD-05
313	BF-60220	BFY-60220	325-01-602-PD-04
314	BF-60206	BFY-60206	325-01-602-PD-08
315	BF-61301	BFY-61301	325-01-613-PD-01
316	BF-61302	BFY-61302	325-01-613-PD-05
317	BF-61303	BFY-61303	325-01-613-PD-01
318	BLM-61302	-	325-01-613-PD-01
319	PD-60202B	PCM-60205-B	325-01-602-PD-01
320	PD-60202C	PCM-60205-C	325-01-602-PD-01
321	PD-60202D	PCM-60205-D	325-01-602-PD-01
322	PD-60202E	PCM-60205-E	325-01-602-PD-01
323	PC-40203	PCY-40201	325-01-402-PD-02
324	PC-40201	PCY-40101	325-01-402-PD-01
325	PC-40202	PCY-40102	325-01-402-PD-01
326	PD-40204		325-01-402-PD-01
327	PD-40205		325-01-402-PD-01
328	PD-60202A	PCY-60205-A	325-01-602-PD-01
329	PCY-60215	-	325-01-602-PD-03
330	BL-61301	PVM-61301	325-01-613-PD-03
331	RVM-60201	-	325-01-602-PD-08
332	RVM-60202	-	325-01-602-PD-08
333	RVM-60203	-	325-01-602-PD-08

334	RVM-60204	-	325-01-602-PD-08
335	RVM-61301	-	325-01-613-PD-01
336	RVM-61302	-	325-01-613-PD-03
337	RVM-61303	-	325-01-613-PD-05
338	RVM-61304	-	325-01-613-PD-01
339	RVM-61305	-	325-01-613-PD-01
340	SCM-61301	-	325-01-613-PD-01
341	SCM-61302	-	325-01-613-PD-01
342	B&B Packing	-	-
343	XA-61301A	-	325-01-613-PD-02
344	XA-61301B	-	325-01-613-PD-02
345	XA-61301C	-	325-01-613-PD-02
346	XA-61301D	-	325-01-613-PD-02
347	XA-61301E	-	325-01-613-PD-02
348	XA-61301F	-	325-01-613-PD-02
349	XA-61301G	-	325-01-613-PD-02
350	XA-61301H	-	325-01-613-PD-02
351	XA-61301J	-	325-01-613-PD-02
352	XS-61301A	-	325-01-613-PD-02
353	XS-61301B	-	325-01-613-PD-02
354	XS-61301C	-	325-01-613-PD-02
355	XS-61301D	-	325-01-613-PD-02
356	XS-61301E	-	325-01-613-PD-02
357	XS-61301F	-	325-01-613-PD-02
358		-	325-01-613-PD-02
359		-	325-01-613-PD-02
360		-	325-01-613-PD-02
361		-	325-01-613-PD-02
362	Boiler	-	-
363	XA-92201B		325-01-922-PD-03
363	XA-92201C		325-01-922-PD-03
363	XA-92212B		325-01-922-PD-03
363	XL-92201B		325-01-922-PD-03
363	XL-92201C		325-01-922-PD-03
363	ZL-92201		325-01-922-PD-03

363		-	325-01-922-PD-03
	Bin Vent Filter (FA-61301)	-	-
363	XA-FA-61301A		325-01-613-PD-01
364	XA-FA-61301B		325-01-613-PD-01
365	XS-FA-61301		325-01-613-PD-01
	Fugitive Dust Filter (FA-61310)	-	-
363	XA-FA-61310A		325-01-613-PD-01
364	XA-FA-61310B		325-01-613-PD-01
365	XS-FA-61310		325-01-613-PD-01
	Vacuum Cleaner Filter (FA-61311)	-	-
363	XA-FA-61311A		325-01-613-PD-03
364	XA-FA-61311B		325-01-613-PD-03
365	XS-FA-61311		325-01-613-PD-03
	Local Exhaust Ventilation Filter (FA-61312)	-	-
363	XA-FA-61312A		325-01-613-PD-05
364	XA-FA-61312B		325-01-613-PD-05
365	XS-FA-61312		325-01-613-PD-05
362	Bag Filter Control Panel (FA-60240)	-	-
363	XS-60240A	-	325-01-602-PD-05
364	XS-60240B	-	325-01-602-PD-05
365	XA-60240A	-	325-01-602-PD-05
366	XA-60240B	-	325-01-602-PD-05
367	DPI-60240	-	325-01-602-PD-05
368	Filter 22 Header Control Panel (TPACINC)	-	-
369	XS-60241A	-	325-01-602-PD-05
370	XA-60241A	-	325-01-602-PD-05
371	XA-60241B	-	325-01-602-PD-05
372	TSH-60241A	-	325-01-602-PD-05
373	TSL-60241B	-	325-01-602-PD-05
374	Filter 22 Header Control Panel (TPACINC)	-	-
375	XS-60242A	-	325-01-602-PD-05
376	XA-60242A	-	325-01-602-PD-05
377	XA-60242B	-	325-01-602-PD-05
378	TSH-60242A	-	325-01-602-PD-05
379	TSL-60242B	-	325-01-602-PD-05

380	THRE-20 Heater Control Panel (TDACINC)	-	-
381	XS-60243A	-	325-01-602-PD-05
382	XA-60243A	-	325-01-602-PD-05
383	XA-60243B	-	325-01-602-PD-05
384	TSH-60243A	-	325-01-602-PD-05
385	TSL-60243B	-	325-01-602-PD-05
386	THRE-20 Heater Control Panel (DK-60210)	-	-
387	XA-FH-60210A	-	325-01-602-PD-06
388	XA-FH-60210B	-	325-01-602-PD-06
389	XA-FH-60210C	-	325-01-602-PD-06
390	XA-FH-60210D	-	325-01-602-PD-06
391	XA-FH-60210E	-	325-01-602-PD-06
392	XS-FH-60210A	-	325-01-602-PD-06
393	XS-FH-60210B	-	325-01-602-PD-06
394	XS-FH-60210C	-	325-01-602-PD-06
395	XS-FH-60210D	-	325-01-602-PD-06
396	XS-FH-60210E		325-01-602-PD-06
397	XS-FH-60210F		325-01-602-PD-06
398	XS-FH-60210G		325-01-602-PD-06
399	TI-FH-60210	-	325-01-602-PD-06
400	UY-FH-60210A	-	325-01-602-PD-06
401	UY-FH-60210B	-	325-01-602-PD-06
402	COM-FH-60210B	-	325-01-602-PD-06
403	TE-60212	-	-
404	DPHL-60230	-	-
405	DPSL-60211	-	-
406	FCV-60211B	-	-
407	ZT-60211B	-	-
408	ZSL-60211B	-	-
409	ZSH-60211B	-	-
410	VJ-01 (24VDC)	-	-
411	VJ-01 (220VAC)	-	-
412	UJ-01 (24VDC)	-	-
413	UJ-01 (220VAC)	-	-
414	PV-60210A	-	-

415	PV-60210B	-	-
416	DPT-60210	-	-
417	DPSL-60210B	-	-
418	DPSH-60210C	-	-
419	BE-60210A	-	-
420	BE-60210B	-	-
421	IX-60210A	-	-
422	FCV-60210A	-	-
423	ZSL-60210A	-	-
424	ZT-60210A	-	-
425	PSH-60210	-	-
426	BV-60210B	-	-
427	ZSO-60210B	-	-
428	ZSC-60210B	-	-
429	VV-60210	-	-
430	ZSO-60210C	-	-
431	ZSC-60210C	-	-
432	PSL-60210	-	-
433	BV-60210A	-	-
434	ZSO-60210A	-	-
435	ZSC-60210A	-	-
436	High Pressure Pump (PD-60202)	-	-
437	XS-60202	ES-60202	325-01-602-PD-01
438	XA-60202		325-01-602-PD-01
439	PT-60202	PT-60205	325-01-602-PD-01
440	FS-60202	FS-60205	325-01-602-PD-01
441	XS-60205	-	325-01-602-PD-01
442	VS-60202	VS-60205	325-01-602-PD-01
443	XV-60202A	XV-60205A	325-01-602-PD-01
444	PS-60202		325-01-602-PD-01
445	TT-60202A		325-01-602-PD-01
446	TT-60202B		325-01-602-PD-01
447	YA-60205	-	325-01-602-PD-01
448	YI-60205	-	325-01-602-PD-01
449	Suppression Panel	-	-

450	PSH-61301	-	325-01-613-PD-01
451	PSH-60220	-	325-01-602-PD-04
452	PSH-60230	-	325-01-602-PD-04
453	PSH-60231	-	325-01-602-PD-04
454	PSH-60224	-	325-01-602-PD-05
455	XA-60202A	-	325-01-602-PD-08
456	XA-60202B	-	325-01-602-PD-08
457	XA-60202C	-	325-01-602-PD-08
458	XA-60202D	-	325-01-602-PD-08
459	XA-60202E	-	325-01-602-PD-08
460	XA-60202F	-	325-01-602-PD-08
461	XA-60202G	-	325-01-602-PD-08
462	XA-60202H	-	325-01-602-PD-08
463	XA-60202J	-	325-01-602-PD-08
464	XA-60202K	-	325-01-602-PD-08
465	XA-60202L	-	325-01-602-PD-08
466	XA-60202M	-	325-01-602-PD-08
467	Fire Deluge PK-60221	-	-
468	XA-60221A	XA-60261	325-01-602-PD-04
469	XA-60221B	XA-60262	325-01-602-PD-04
470	XS-60221	XS-60260	325-01-602-PD-04
139	TSH-60230		325-01-602-PD-04
471	Fire Deluge PK-60222	-	-
472	XA-60222A		325-01-602-PD-05
473	XA-60222B		325-01-602-PD-05
474	XS-60222		325-01-602-PD-05
	XV-60240D		325-01-602-PD-05
140	TSH-60240		325-01-602-PD-05
475	Keaser Air Blower	-	-
476	XS-61302		325-01-613-PD-01
477	XA-61302A		325-01-613-PD-01
478	XA-61302B		325-01-613-PD-01
479	XA-61302C		325-01-613-PD-01
480	XA-61302D		325-01-613-PD-01
481	REMOTE I-O Area-200	-	-

482	HV-20204	HV-2022	325-01-202-PD-01
483	HV-20205	HV-2024	325-01-202-PD-01
484	HV-20206	HV-2026	325-01-202-PD-01
485	HV-20207	HV-2030	325-01-202-PD-01
486	HV-20208	HV-2032	325-01-202-PD-02
487	HV-20209	HV-2035	325-01-202-PD-02
488	HV-20210	HV-2038	325-01-202-PD-02
489	XV-20201	-	325-01-202-PD-01
490	XV-20202	-	325-01-202-PD-01
491	XV-20203	-	325-01-202-PD-01
492	XV-20211	-	325-01-202-PD-01
493	XV-20212	-	325-01-202-PD-01
494	XV-20213	-	325-01-202-PD-01
495	LS-814201	LS-81001	325-01-814-PD-01
496	PC-20201	PCM-2030	325-01-202-PD-01
497	PCM-2031	-	325-01-202-PD-01
498	PC-814201	PCY-81001	325-01-814-PD-01
499	PC-814202	PCY-81002	325-01-814-PD-01
500	LPG Station	-	-
501	FT-913201	FT-80301	325-01-913-PD-02
502	LT-913201	LT-80301	325-01-913-PD-01
503	PT-913201	PT-80302	325-01-913-PD-02
504	XV-913201		325-01-913-PD-02
505	Gas Vaporizer 4	-	-
506	XA-913201A	XA-80305A	325-01-913-PD-02
507	XA-913201B	XA-80305B	325-01-913-PD-02
508	XA-913201C	XA-80305C	325-01-913-PD-02
509	XA-913201D	XA-80305D	325-01-913-PD-02
510	XA-913201E	XA-80305E	325-01-913-PD-02
512	XS-913201A	XS-80305A	325-01-913-PD-02
512	XS-913201B	XS-80305B	325-01-913-PD-02
513	Gas Vaporizer 5	-	-
514	XA-913202A	XA-80306A	325-01-913-PD-02
515	XA-913202B	XA-80306B	325-01-913-PD-02
516	XA-913202C	XA-80306C	325-01-913-PD-02

517	XA-913202D	XA-80306D	325-01-913-PD-02
518	XA-913202E	XA-80306E	325-01-913-PD-02
519	XS-913202A	XS-80306A	325-01-913-PD-02
520	XS-913202B	XS-80306B	325-01-913-PD-02

Instrument Service Description	Instrument Type / Name	Signal Type
Dust Analyzer Transmitter for Exhaust Duct (DTEX-22)	Dust Analyzer Transmitter	4 - 20 mA
Steam filter pressure	Differential Pressure Transmitter	4 - 20 mA
Vent filter pressure	Differential Pressure Transmitter	4 - 20 mA
Vent filter pressure	Differential Pressure Transmitter	4 - 20 mA
Vent filter pressure	Differential Pressure Transmitter	4 - 20 mA
Vent filter pressure	Differential Pressure Transmitter	4 - 20 mA
Gas Pressure Differential Transmitter for Air Filter (FICA-31)	Differential Pressure Transmitter	4 - 20 mA
Gas Pressure Differential Transmitter for Air Filter (FICA-21)	Differential Pressure Transmitter	4 - 20 mA
Slurry low flow	Flow Switch	Dry Contact
Slurry low flow	Flow Switch	Dry Contact
Slurry low flow	Flow Switch	Dry Contact
Jet Cooker	Flow Switch	Dry Contact
Jet Cooker	Flow Switch	Dry Contact
Slurry flowrate & density check transfer line	Coriolis Flow Transmitter	4 - 20 mA,Pulse
Feed Slurry Flow Transmitter for Pipe (PIPE)	Magnetic Flow Transmitter	4 - 20 mA,Pulse
Jet Cooker	Coriolis Flow Transmitter	4 - 20 mA,Pulse
Jet Cooker	Coriolis Flow Transmitter	4 - 20 mA,Pulse
Cool air Flow	Thermal Mass Flow Transmitter	4 - 20 mA
Gas Flow Transmitter for Cooling Conveying Cold Air Duct (DTCA-32)	Thermal Mass Flow Transmitter	4 - 20 mA
Gas Flow Element for Dryer Hot Air Duct (DTHA-21B)	Thermal Mass Flow Transmitter	4 - 20 mA
Gas Flow Element for Dryer Hot Air Duct (DTHA-21A)	Thermal Mass Flow Transmitter	4 - 20 mA
Feed Slurry Flow Transmitter for Nozzle #1 (AMTV-11A)	Ultrasonic Flow Transmitter	4 - 20 mA
Feed Slurry Flow Transmitter for Nozzle #2 (AMTV-11A)	Ultrasonic Flow Transmitter	4 - 20 mA
Feed Slurry Flow Transmitter for Nozzle #3 (AMTV-11A)	Ultrasonic Flow Transmitter	4 - 20 mA
Feed Slurry Flow Transmitter for Nozzle #4 (AMTV-11A)	Ultrasonic Flow Transmitter	4 - 20 mA
Feed Slurry Flow Transmitter for Nozzle #5 (AMTV-11B)	Ultrasonic Flow Transmitter	4 - 20 mA
Feed Slurry Flow Transmitter for Nozzle #6 (AMTV-11B)	Ultrasonic Flow Transmitter	4 - 20 mA
Feed Slurry Flow Transmitter for Nozzle #7 (AMTV-11B)	Ultrasonic Flow Transmitter	4 - 20 mA
Feed Slurry Flow Transmitter for Nozzle #8 (AMTV-11B)	Ultrasonic Flow Transmitter	4 - 20 mA
Feed Slurry Flow Transmitter for Nozzle #9 (AMTV-11C)	Ultrasonic Flow Transmitter	4 - 20 mA
Feed Slurry Flow Transmitter for Nozzle #10 (AMTV-11C)	Ultrasonic Flow Transmitter	4 - 20 mA

Feed Slurry Flow Transmitter for Nozzle #11 (AMTV-11C)	Ultrasonic Flow Transmitter	4 - 20 mA
Feed Slurry Flow Transmitter for Nozzle #12 (AMTV-11C)	Ultrasonic Flow Transmitter	4 - 20 mA
Feed Slurry Flow Transmitter for Nozzle #13 (AMTV-11D)	Ultrasonic Flow Transmitter	4 - 20 mA
Feed Slurry Flow Transmitter for Nozzle #14 (AMTV-11D)	Ultrasonic Flow Transmitter	4 - 20 mA
Feed Slurry Flow Transmitter for Nozzle #15 (AMTV-11D)	Ultrasonic Flow Transmitter	4 - 20 mA
Feed Slurry Flow Transmitter for Nozzle #16 (AMTV-11D)	Ultrasonic Flow Transmitter	4 - 20 mA
Compressed air Flow	Thermal Mass Flow Transmitter	4 - 20 mA
Culinary steam feed to Spray dryer	Vortex Flow Transmitter	4 - 20 mA
Steam feed to Spray dryer	Vortex Flow Transmitter	4 - 20 mA
Compressed air Flow	Thermal Mass Flow Transmitter	4 - 20 mA
Process Water	Magnetic Flow Transmitter	4 - 20 mA,Pulse
Process Water	Magnetic Flow Transmitter	4 - 20 mA,Pulse
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Start/Stop	Hand Switch	Dry Contact
Slurry Drain tank TS-40201	Valve With Position Switch	Dry Contact
Slurry Drain line	Valve With Position Switch	Dry Contact
Slurry Drain tank TS-40202	Valve With Position Switch	Dry Contact
Slurry Drain line	Valve With Position Switch	Dry Contact
Slurry Drain tank TS-40203	Valve With Position Switch	Dry Contact

Slurry Drain line	Valve With Position Switch	Dry Contact
Jet Cooker	Valve With Position Switch	Dry Contact
Jet Cooker	Valve With Position Switch	Dry Contact
Powder Level Switch High for Cyclone (CYCL-21)	Level switch	Dry Contact
Powder Level Switch High for Cyclone (CYCL-22)	Level switch	Dry Contact
Powder Level Switch High for Bag House (BGHS-21)	Level switch	Dry Contact
Water Level Switch High for Fire Deluge Pipe (FWDP-21)	Level switch	Dry Contact
Water Level Switch High for Fire Deluge Pipe (FWDP-22)	Level switch	Dry Contact
Level low switch	Level switch	Dry Contact
Level alarm high	Level switch	Dry Contact
Level alarm low	Level switch	Dry Contact
Level switch	Level switch	Dry Contact
Big bag level alarm high	Level switch	Dry Contact
Big bag level alarm high	Level switch	Dry Contact
Level alarm high	Level switch	Dry Contact
High Level alarm reject screen	Level switch	Dry Contact
High Level alarm Cooling cyclone	Level switch	Dry Contact
Level alarm high	Level switch	Dry Contact
Low Level alarm		Dry Contact
Level transmitter in pH Adjustment tank 1	Level transmitter	4 - 20 mA
Level transmitter in pH Adjustment tank 2	Level transmitter	4 - 20 mA
Level transmitter in Spray dryer infeed tank	Level transmitter	4 - 20 mA
Level powder in silo	Level transmitter	4 - 20 mA
pH analyzer in pH Adjustment tank 1	pH Analyzer	4 - 20 mA
pH analyzer in pH Adjustment tank 2	pH Analyzer	4 - 20 mA
Gas Pressure Switch Low for Spray Chamber (CHAM-21)	Pressure switch	Dry Contact
Pressure alarm low	Pressure switch	Dry Contact
Pressure alarm low	Pressure switch	Dry Contact
Pressure alarm low	Pressure switch	Dry Contact
Pressure alarm low	Pressure switch	Dry Contact
Pressure alarm low	Pressure switch	Dry Contact
Pressure Transmitter for Inlet Filter Strainer (FSTR-11)	Pressure Transmitter	4 - 20 mA
Feed Slurry Pressure Transmitter for Pipe (PIPE)	Pressure Transmitter	4 - 20 mA
Jet Cooker	Pressure Transmitter	4 - 20 mA
Steam feed to Spray dryer	Pressure Transmitter	4 - 20 mA

Feed Slurry Pressure Transmitter for High Pressure Pump Outlet (PUPD-11)	Pressure Transmitter	4 - 20 mA
Gas Pressure Transmitter for Cool Air Duct (DTCA-32)	Pressure Transmitter	4 - 20 mA
Gas Pressure Transmitter for Hot Air Duct (DTHA-21)	Pressure Transmitter	4 - 20 mA
15% Polypropylene Glycol Pressure Indicator Transmitter for Heat Recovery Pumping Skid (DPICE-21)	Pressure Transmitter	4 - 20 mA
Gas Pressure Transmitter for Combustion Air Burner Duct (DTCB-21)	Pressure Transmitter	4 - 20 mA
Gas Pressure Transmitter for Chamber (CHAM-21)	Pressure Transmitter	4 - 20 mA
Gas Pressure Transmitter for Cooling Air Duct (DTCA-25)	Pressure Transmitter	4 - 20 mA
Gas Pressure Transmitter for Conveying Duct (DTCO-21)	Pressure Transmitter	4 - 20 mA
Gas Pressure Transmitter for Cyclone Exhaust Duct (DTCE-21)	Pressure Transmitter	4 - 20 mA
Gas Pressure Transmitter for Bag House Exhaust Duct (DTEX-22)	Pressure Transmitter	4 - 20 mA
Gas Pressure Transmitter for Exhaust Fan (FANE-21)	Pressure Transmitter	4 - 20 mA
Pressure transmitter	Pressure Transmitter	4 - 20 mA
Pressure in silo	Pressure Transmitter	4 - 20 mA
Pressure in vacuum line	Pressure Transmitter	4 - 20 mA
Powder Pressure Transmitter for Cooling / Conveyor Cyclone (CYCL-31)	Pressure Transmitter	4 - 20 mA
Gas Pressure Transmitter for Storage Bin Cyclone Inlet Duct (CYCL-31)	Pressure Transmitter	4 - 20 mA
Gas Pressure Transmitter for Exhaust Duct (DTEX-31)	Pressure Transmitter	4 - 20 mA
Pressure transmitter	Pressure Transmitter	4 - 20 mA
Steam feed to Spray dryer	Automatic Control valve	4 - 20 mA
Water service pH analyzer in pH Adjustment tank 1	Solenoid valve	24 VDC energized
Water service pH analyzer in pH Adjustment tank 2	Solenoid valve	24 VDC energized
Air service pH analyzer in pH Adjustment tank 1	Solenoid valve	24 VDC energized
Air service pH analyzer in pH Adjustment tank 2	Solenoid valve	24 VDC energized
PA On/Off Valve for Chamber PSE purge air (CHAM-21)	Solenoid valve	24 VDC energized
PA On/Off Valve for Cyclone Air Hammer 27 (CYCL-21)	Solenoid valve	24 VDC energized
PA On/Off Valve for Cyclone Air Hammer 28 (CYCL-22)	Solenoid valve	24 VDC energized
PA On/Off Valve for Cyclone Air Hammer 29A (BGHS-21)	Solenoid valve	24 VDC energized
PA On/Off Valve for Cyclone Air Hammer 29B (BGHS-21)	Solenoid valve	24 VDC energized
PA On/Off Valve for Cyclone Air Hammer 29C (BGHS-21)	Solenoid valve	24 VDC energized
Timer compress air pluse solenoid valve	Solenoid valve	24 VDC energized
Air service	Solenoid valve	24 VDC energized
Air service	Solenoid valve	24 VDC energized
Timer compress air pluse solenoid valve	Solenoid valve	24 VDC energized
Air service	Solenoid valve	24 VDC energized
Timer compress air pluse solenoid valve	Solenoid valve	24 VDC energized

Air service	Solenoid valve	24 VDC energized
Timer compress air pluse solenoid valve	Solenoid valve	24 VDC energized
Air service	Solenoid valve	24 VDC energized
Gas and Powder Temperature Switch High for Conveying Duct (DTCO-21)	Temperature Switch	Dry Contact
Gas and Powder Temperature Switch High for Exhaust Duct (DTEX-22)	Temperature Switch	Dry Contact
Jet Cooker	Temperature Transmitter	4 - 20 mA
Jet Cooker	Temperature Transmitter	4 - 20 mA
Steam feed to Spray dryer	Temperature Transmitter	4 - 20 mA
Gas Temperature Element for Hot Air Duct (DTHA-21)	Temperature Transmitter	4 - 20 mA
Bearing Temperature Transmitter for Main Supply Air Fan (FANS-21)	Temperature Transmitter	4 - 20 mA
Bearing Temperature Transmitter for Main Supply Air Fan (FANS-21)	Temperature Transmitter	4 - 20 mA
Gas Temperature Element for Hot Air Duct (DTHA-21)	Temperature Transmitter	4 - 20 mA
Steam Temperature Transmitter for Cold Air Duct (DTCA-23)	Temperature Transmitter	4 - 20 mA
Liquid Temperature Transmitter for Heat Recovery (HXCR-21)	Temperature Transmitter	4 - 20 mA
Liquid Temperature Transmitter for Heat Recovery (HXCR-21)	Temperature Transmitter	4 - 20 mA
Gas Temperature Transmitter for Combustion Air Burner Duct (DTCB-21)	Temperature Transmitter	4 - 20 mA
Gas and Powder Temperature Transmitter for Spray Chamber (CHAM-21)	Temperature Transmitter	4 - 20 mA
Gas and Powder Temperature Transmitter for Spray Chamber (CHAM-21)	Temperature Transmitter	4 - 20 mA
Gas and Powder Temperature Transmitter for Spray Chamber (CHAM-21)	Temperature Transmitter	4 - 20 mA
Gas and Powder Temperature Transmitter for Spray Chamber (CHAM-21)	Temperature Transmitter	4 - 20 mA
Gas and Powder Temperature Transmitter for Spray Chamber (CHAM-21)	Temperature Transmitter	4 - 20 mA
Gas and Powder Temperature Transmitter for Spray Chamber (CHAM-21)	Temperature Transmitter	4 - 20 mA
Gas and Powder Temperature Transmitter for Spray Chamber (CHAM-21)	Temperature Transmitter	4 - 20 mA
Gas and Powder Temperature Transmitter for Spray Chamber (CHAM-21)	Temperature Transmitter	4 - 20 mA
Gas and Powder Temperature Transmitter for Spray Chamber (CHAM-21)	Temperature Transmitter	4 - 20 mA
Gas and Powder Temperature Transmitter for Conveying Duct (DTCO-21)	Temperature Transmitter	4 - 20 mA
Gas Temperature Transmitter for Exhaust Fan (FANE-21)	Temperature Transmitter	4 - 20 mA
Bearing Temperature Transmitter for Main Exhaust Air Fan (FANE-21)	Temperature Transmitter	4 - 20 mA
Bearing Temperature Transmitter for Main Exhaust Air Fan (FANE-21)	Temperature Transmitter	4 - 20 mA
Gas Temperature Transmitter for Exhaust Stack (DTXS-21)	Temperature Transmitter	4 - 20 mA
Compress air temp to camera	Temperature Transmitter	4 - 20 mA
Gas Temperature Transmitter for Cold Air Duct (DTCA-32)	Temperature Transmitter	4 - 20 mA
Gas and Powder Temperature Transmitter for Cooling Cyclone (CYCL-31)	Temperature Transmitter	4 - 20 mA
Jet Cooker	Automatic Control valve	4 - 20 mA
Steam Temperature Valve for Pre-Heat Steam Heat Exchanger (HXCH-21)	Automatic Control valve	4 - 20 mA
Chiller water control valve	3-way Control valve	4 - 20 mA

[illegible]

Jet Cooker	Automatic on/off valve	24 VDC energized
Jet Cooker	Automatic on/off valve	24 VDC energized
Steam feed to Spray dryer	Automatic on/off valve	24 VDC energized
Liquid On/Off Valve for Drain Pipe (PIPE)	Automatic on/off valve	24 VDC energized
Steam feed to Spray dryer	Automatic on/off valve	24 VDC energized
High Pressure Water On/Off Valve for Drain Pipe	Automatic on/off valve	24 VDC energized
Product feed-out Packing silo	Automatic on/off valve	24 VDC energized
Dust collector feed vent	Automatic on/off valve	24 VDC energized
Dust collector feed from packer	Automatic on/off valve	24 VDC energized
On/Off Valve for Diverter Valve DV-61301	Automatic on/off valve	24 VDC energized
On/Off Valve for Diverter Valve DV-61302	Automatic on/off valve	24 VDC energized
Limit Switch for CHAM-21 (Spray Chamber)	Limit Switch	Dry Contact
Limit Switch for CHAM-21 (Spray Chamber)	Limit Switch	Dry Contact
Limit Switch for CHAM-21 (Spray Chamber)	Limit Switch	Dry Contact
Limit Switch for CHAM-21 (Spray Chamber)	Limit Switch	Dry Contact
Limit Switch for CHAM-21 (Spray Chamber)	Limit Switch	Dry Contact
Limit Switch for CY-60230 (Door Service)	Limit Switch	Switch PNP
Limit Switch for CY-60231 (Door Service)	Limit Switch	Switch PNP
Gas Limit Switch for DTEX-21 (Bag House Face Spectacle Inserted)	Limit Switch	Dry Contact
Gas Limit Switch for DTEX-21 (Bag House Face Blind Inserted)	Limit Switch	Dry Contact
Gas Limit Switch for DTEX-23 (Bag House By-Pass Inlet Spool Inserted)	Limit Switch	Switch PNP
Gas Limit Switch for DTEX-23 (Bag House By-Pass Inlet Blind Inserted)	Limit Switch	Switch PNP
Gas Limit Switch for DTEX-23 (Bag House By-Pass Outlet Spool Inserted)	Limit Switch	Switch PNP
Gas Limit Switch for DTEX-23 (Bag House By-Pass Outlet Blind Inserted)	Limit Switch	Switch PNP
Limit Switch for FA-60240 (Door Service)	Limit Switch	Switch PNP
Position Magnet Door	Limit Switch	Dry Contact
Position Magnet Door	Limit Switch	Dry Contact
Position Magnet Door	Limit Switch	Dry Contact
Position Magnet Door	Limit Switch	Dry Contact
Lototo-01	Position switch	Dry Contact
Lototo-02	Position switch	Dry Contact
Lototo-03	Position switch	Dry Contact
Lototo-04	Position switch	Dry Contact
Lototo-05	Position switch	Dry Contact
Lototo-06	Position switch	Dry Contact

Lototo-07	Position switch	Dry Contact
Lototo-08	Position switch	Dry Contact
Lototo-09	Position switch	Dry Contact
Lototo-10	Position switch	Dry Contact
Lototo-11	Position switch	Dry Contact
Lototo-12	Position switch	Dry Contact
Emergency stop activated @ Spray Nozzle 7th Floor	E-Stop Switch	Dry Contact
Emergency stop activated @ Sight glass 6th Floor	E-Stop Switch	Dry Contact
Emergency stop activated @ Control Room	E-Stop Switch	Dry Contact
Emergency stop activated @ SC-61301,SC-61302	E-Stop Switch	Dry Contact
		-
Instr. Air Flow Transmitter for Camera #4 (ACAM-21)	Flow Transmitter	4 - 20 mA
Instr. Air Pressure Transmitter for Camera #4 (ACAM-21)	Pressure Transmitter	4 - 20 mA
PA On/Off Valve for Bag House Tank Valve (BGHS-21)	On/Off Valve	24 VDC energized,Dry Contact
Instr. Air On/Off Valve for Camera #4 (ACAM-21)	On/Off Valve	24 VDC energized
PA On/Off Valve for Cyclone-21 Rotary Valve (RVAL-21)	On/Off Valve	24 VDC energized
Speed Switch Low for Bin Cyclone Rotary Valve (RVAL-21)	Low Speed Switch	Switch PNP
Position - Rotary Valve (RVAL-21/ RV-60201) Casing Closed	Position switch	Dry Contact
PA On/Off Valve for Cyclone-22 Rotary Valve (RVAL-22)	On/Off Valve	24 VDC energized
Speed Switch Low for Bin Cyclone Rotary Valve (RVAL-22)	Low Speed Switch	Switch PNP
Position - Rotary Valve (RVAL-22/ RV-60202) Casing Closed	Position switch	Dry Contact
PA On/Off Valve for Bag House Rotary Valve (RVAL-23)	On/Off Valve	24 VDC energized
Speed Switch Low for Bin Cyclone Rotary Valve (RVAL-23)	Low Speed Switch	Switch PNP
Position - Rotary Valve (RVAL-23/ RV-60203) Casing Closed	Position switch	Dry Contact
PA On/Off Valve for Conveying Cooler Cyclone-31 Rotary Valve (RV-61304)	On/Off Valve	24 VDC energized
Speed Switch Low for Bin Cyclone Rotary Valve (RVAL-31)	Low Speed Switch	Switch PNP
Position - Rotary Valve (RVAL-31/ RV-61304) Casing Closed	Position switch	Dry Contact
Instr. Air Temperature Transmitter for Air Manifold - Camera #4 (ACAM-21)	Temperature Transmitter	4 - 20 mA
Feed Slurry On/Off Valve for Nozzle #1 (ATVM-11A)	On/Off Valve	24 VDC energized,Dry Contact
Feed Slurry On/Off Valve for Nozzle #2 (ATVM-11A)	On/Off Valve	24 VDC energized,Dry Contact
Feed Slurry On/Off Valve for Nozzle #3 (ATVM-11A)	On/Off Valve	24 VDC energized,Dry Contact
Feed Slurry On/Off Valve for Nozzle #4 (ATVM-11A)	On/Off Valve	24 VDC energized,Dry Contact
Steam On/Off Valve for Nozzle #1 (ATVM-11A)	On/Off Valve	24 VDC energized,Dry Contact
Steam On/Off Valve for Nozzle #2 (ATVM-11A)	On/Off Valve	24 VDC energized,Dry Contact
Steam On/Off Valve for Nozzle #3 (ATVM-11A)	On/Off Valve	24 VDC energized,Dry Contact

Steam On/Off Valve for Nozzle #4 (ATVM-11A)	On/Off Valve	energized,Dry Հարժաժ
Feed Slurry On/Off Valve for Nozzle #5 (ATVM-11B)	On/Off Valve	energized,Dry Հարժաժ
Feed Slurry On/Off Valve for Nozzle #6 (ATVM-11B)	On/Off Valve	energized,Dry Հարժաժ
Feed Slurry On/Off Valve for Nozzle #7 (ATVM-11B)	On/Off Valve	energized,Dry Հարժաժ
Feed Slurry On/Off Valve for Nozzle #8 (ATVM-11B)	On/Off Valve	energized,Dry Հարժաժ
Steam On/Off Valve for Nozzle #5 (ATVM-11B)	On/Off Valve	energized,Dry Հարժաժ
Steam On/Off Valve for Nozzle #6 (ATVM-11B)	On/Off Valve	energized,Dry Հարժաժ
Steam On/Off Valve for Nozzle #7 (ATVM-11B)	On/Off Valve	energized,Dry Հարժաժ
Steam On/Off Valve for Nozzle #8 (ATVM-11B)	On/Off Valve	energized,Dry Հարժաժ
Feed Slurry On/Off Valve for Nozzle #9 (ATVM-11C)	On/Off Valve	energized,Dry Հարժաժ
Feed Slurry On/Off Valve for Nozzle #10 (ATVM-11C)	On/Off Valve	energized,Dry Հարժաժ
Feed Slurry On/Off Valve for Nozzle #11 (ATVM-11C)	On/Off Valve	energized,Dry Հարժաժ
Feed Slurry On/Off Valve for Nozzle #12 (ATVM-11C)	On/Off Valve	energized,Dry Հարժաժ
Steam On/Off Valve for Nozzle #9 (ATVM-11C)	On/Off Valve	energized,Dry Հարժաժ
Steam On/Off Valve for Nozzle #10 (ATVM-11C)	On/Off Valve	energized,Dry Հարժաժ
Steam On/Off Valve for Nozzle #11 (ATVM-11C)	On/Off Valve	energized,Dry Հարժաժ
Steam On/Off Valve for Nozzle #12 (ATVM-11C)	On/Off Valve	energized,Dry Հարժաժ
Feed Slurry On/Off Valve for Nozzle #13 (ATVM-11D)	On/Off Valve	energized,Dry Հարժաժ
Feed Slurry On/Off Valve for Nozzle #14 (ATVM-11D)	On/Off Valve	energized,Dry Հարժաժ
Feed Slurry On/Off Valve for Nozzle #15 (ATVM-11D)	On/Off Valve	energized,Dry Հարժաժ
Feed Slurry On/Off Valve for Nozzle #16 (ATVM-11D)	On/Off Valve	energized,Dry Հարժաժ
Steam On/Off Valve for Nozzle #13 (ATVM-11D)	On/Off Valve	energized,Dry Հարժաժ
Steam On/Off Valve for Nozzle #14 (ATVM-11D)	On/Off Valve	energized,Dry Հարժաժ
Steam On/Off Valve for Nozzle #15 (ATVM-11D)	On/Off Valve	energized,Dry Հարժաժ
Steam On/Off Valve for Nozzle #16 (ATVM-11D)	On/Off Valve	energized,Dry Հարժաժ
Slurry Drain On/Off Valve for ATVM-11A	On/Off Valve	energized,Dry Հարժաժ
Slurry Drain On/Off Valve for ATVM-11B	On/Off Valve	energized,Dry Հարժաժ
Slurry Drain On/Off Valve for ATVM-11C	On/Off Valve	energized,Dry Հարժաժ
Slurry Drain On/Off Valve for ATVM-11D	On/Off Valve	energized,Dry Հարժաժ
Gas On/Off Valve for Bag House Isolation Damper (DMPR-22)	On/Off Valve	energized,Dry Contact
Powder On/Off Valve for Diverter Valve (DVTR-21)	On/Off Valve	24 VDC energized
		-
Explosion Vent PSE-61300	Position switch	Dry Contact
Explosion Vent PSE-61301	Position switch	Dry Contact
Explosion isolation flap valve ISV-61302	Position switch	Dry Contact

Explosion Vent PSE-61302	Position switch	Dry Contact
Explosion Vent PSE-61303	Position switch	Dry Contact
Explosion Vent PSE-61304	Position switch	Dry Contact
Position Brether Valve High Pressure PSV-61301	Position switch	Dry Contact
Position Brether Valve Vacuum Pressure PSV-61301	Position switch	Dry Contact
Explosion isolation flap valve ISV-61303	Position switch	Dry Contact
Explosion isolation flap valve ISV-61304	Position switch	Dry Contact
		-
Start / Stop Command for VSD AT-40201		24 VDC energized
Start / Stop Command for VSD AT-40202		24 VDC energized
Start / Stop Command for VSD AT-40203		24 VDC energized
Start / Stop Command for VSD FANS-21 (Main Supply Air Fan)		24 VDC energized
Start / Stop Command for VSD FANC-21 (Combustion Fan)		24 VDC energized
Start / Stop Command for VSD FANE-21 (Exhaust Fan)		24 VDC energized
Start / Stop Command for VSD FANS-22 (Cooling Fan)		24 VDC energized
Start / Stop Command for VSD FANS-31 (Cooling / Conveying Fan)		24 VDC energized
Start / Stop Command for VSD BF-61301 (Packing silo vent fan)		24 VDC energized
Start / Stop Command for VSD BF-61302 (LEV Blower fan)		24 VDC energized
Start / Stop Command for VSD BF-61303 (Fugitive Dust Fan)		24 VDC energized
Start / Stop Command for FVNR BL-61301 (Bin Actuator Blower)	Feeder	Power
Start / Stop Command for FVNR PUPD-11 (Oil Pump)		24 VDC energized
Start / Stop Command for FVNR PUPD-11 (Machine Fan 1)		24 VDC energized
Start / Stop Command for FVNR PUPD-11 (Machine Fan 2)		24 VDC energized
Start / Stop Command for FVNR PUPD-11 (Main motor Fan)		24 VDC energized
Start / Stop Command for VSD PUCF-11 (Centrifugal Feed Pump)		24 VDC energized
Start / Stop Command for VSD PC-40201		24 VDC energized
Start / Stop Command for VSD PC-40202		24 VDC energized
Start / Stop Command for VSD PD-40204		24 VDC energized
Start / Stop Command for VSD PD-40205		24 VDC energized
Start / Stop Command for VSD PUPD-11 (Positive Displacement Pump)		24 VDC energized
Start / Stop Command for VSD PUCF-21 (Heat Recovery Centrifugal Pump)		24 VDC energized
Start / Stop Command for FVNR PV-61301 (Vacuum Cleaning Fan)		24 VDC energized
Start / Stop Command for FVNR RVAL-21 (Cyclone-21 Rotary Valve)		24 VDC energized
Start / Stop Command for FVNR RVAL-22 (Cyclone-22 Rotary Valve)		24 VDC energized
Start / Stop Command for FVNR RVAL-23 (Bag House Rotary Valve)		24 VDC energized

Start / Stop Command for FVNR RV-60204 (Recycle silo Rotary Valve)		24 VDC energized
Start / Stop Command for FVNR RV-61301		24 VDC energized
Start / Stop Command for FVNR RV-61302		24 VDC energized
Start / Stop Command for FVNR RV-61303		24 VDC energized
Start / Stop Command for FVNR RV-61304		24 VDC energized
Start / Stop Command for FVNR RV-61305		24 VDC energized
Start / Stop Command for FVNR SC-61301 (Sifter)		24 VDC energized
Start / Stop Command for FVNR SC-61302 (Sifter)		24 VDC energized
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Request filter plant (Fugitive)	Request filter plant	Dry Contact
Request material transport	Request material transport	Dry Contact
Request belt plant	Request belt plant	Dry Contact
Machine in operation	Machine in operation	Dry Contact
General Fault	General Fault	Dry Contact
Machine is ready	Machine is ready	Dry Contact
Emergency stop activated	Emergency stop activated	Dry Contact
Silo minimum	Silo minimum	Dry Contact
Silo maximum	Silo maximum	Dry Contact
Filter plant in operation	Filter plant in operation	24 VDC energized
Material transport in operation	Material transport in operation	24 VDC energized
Belt plant in operation	Belt plant in operation	24 VDC energized
Sequence Stop	Sequence Stop	24 VDC energized
Status from Customer Emergency stop	Status from Customer Emergency stop	24 VDC energized
Operation Release	Operation Release	24 VDC energized
Spare		Dry Contact
Spare		24 VDC energized
Spare		24 VDC energized
Communication Signal		Communication Signal
		-
Boiler common alarm		Dry Contact
Boiler Trip		Dry Contact
Burner Trip		Dry Contact
Boiler Operate		Dry Contact
Remote Setpoint Activated		Dry Contact
Remote Setpoint Status		Dry Contact

Communication Signal		Communication Signal
		-
Start / Stop Command for FA-61301		Dry Contact
Compress air low (Spare)		Dry Contact
Alarm 1		Dry Contact
		-
Start / Stop Command for FA-61310		Dry Contact
Compress air low (Spare)		Dry Contact
Alarm 1		Dry Contact
		-
Start / Stop Command for FA-61311		Dry Contact
Compress air low (Spare)		Dry Contact
Alarm 1		Dry Contact
		-
Start / Stop Command for FA-61311		Dry Contact
Compress air low (Spare)		Dry Contact
Alarm 1		Dry Contact
		-
Start / Stop Command for BGHS-21 (Bag House Air Pulsing)		24 VDC energized
Fan Command for BGHS-21 (Bag House Air Pulsing)		24 VDC energized
Alarm 1 for BGHS-21 (Bag House Air Pulsing)		Dry Contact
Alarm 2 for BGHS-21 (Bag House Air Pulsing)		Dry Contact
Gas Pressure Differential Pressure (Bag House Air Pulsing)		4 - 20 mA
		-
Start / Stop Command for HTRE-21 (Heater Control Panel)		24 VDC energized
Alarm 1 for BGHS-21 (Bag House Air Pulsing)		Dry Contact
Alarm 2 for BGHS-21 (Bag House Air Pulsing)		Dry Contact
Surface Heating Temperature switch high for bag house cone (BGHS-21)	Temperature Switch	Dry Contact
Surface Heating Temperature switch low for bag house cone (BGHS-21)	Temperature Switch	Dry Contact
		-
Start / Stop Command for HTRE-21 (Heater Control Panel)		24 VDC energized
Alarm 1 for BGHS-21 (Bag House Air Pulsing)		Dry Contact
Alarm 2 for BGHS-21 (Bag House Air Pulsing)		Dry Contact
Surface Heating Temperature switch high for bag house cone (BGHS-21)	Temperature Switch	Dry Contact
Surface Heating Temperature switch low for bag house cone (BGHS-21)	Temperature Switch	Dry Contact

		-
Start / Stop Command for HTRE-21 (Heater Control Panel)		24 VDC energized
Alarm 1 for BGHS-21 (Bag House Air Pulsing)		Dry Contact
Alarm 2 for BGHS-21 (Bag House Air Pulsing)		Dry Contact
Surface Heating Temperature switch high for bag house cone (BGHS-21)	Temperature Switch	Dry Contact
Surface Heating Temperature switch low for bag house cone (BGHS-21)	Temperature Switch	Dry Contact
		-
BMS PANEL POWER ON for AHTR-21		Dry Contact
BMS Burner High Fire Purge for AHTR-21		Dry Contact
BMS Burner On for AHTR-21		Dry Contact
BMS BMS Lockout for AHTR-21		Dry Contact
BMS Remote Modet for AHTR-21		Dry Contact
Burner Remote Start for AHTR-21		24 VDC energized
Dryer Interlocks for AHTR-21		24 VDC energized
Remote Reset for AHTR-21		24 VDC energized
Spare		24 VDC energized
Exhaust fan run status BF-60240		Dry Contact
Supply fan run status BF-60210		Dry Contact
Combustion fan run status BF-60211		Dry Contact
Gas Burner Firing Rate Temp Signal		4 - 20 mA
Comb fan VFD Speed Control		4 - 20 mA
Demand Signal		4 - 20 mA
Communication Signal		Modbus RTU
Gas Temperature Element for Hot Air Duct (DTHA-21)	Temperature Element	T/C Type K
Gas Pressure Differential Switch Low for Exhaust Fan (FANE-21)	Pressure Differntial Switch Low	Dry Contact
Gas Pressure Differential Switch Low for Combustion Fan (FANC-21)	Pressure Differential Switch Low	Dry Contact
Gas Flow Control Damper (Valve) for Burner Combustion Air Duct (DTCB-21)	Flow Control Damper (Valve)	Internal wiring
Gas Position Transmitter for Combustion Air Damper Actuator (DTCB-21)	Position Transmitter	Internal wiring
Gas Position Switch Low for Combustion Air Damper Actuator (DTCB-21)	Position Switch Low	Internal wiring
Gas Position Switch High for Combustion Air Damper Actuator (DTCB-21)	Position Switch High	Internal wiring
Valve train J-BOX	Control Box	Internal wiring
Valve train J-BOX	Control Box	Internal wiring
Valve train J-BOX	Control Box	Internal wiring
Valve train J-BOX	Control Box	Internal wiring
Gas Pressure Valve for Air Heater Gas Train (GTRN-21)	Pressure Valve	Internal wiring

Gas Pressure Valve for Air Heater Gas Train (GTRN-21)	Pressure Valve	Internal wiring
Gas Pressure Differential Switch Low for Main Supply Air Fan (FANS-21)	Pressure Differential Switch Low	Internal wiring
Gas Pressure Differential Switch Low for Air Heater (AHTR-21)	Pressure Differential Switch Low	Internal wiring
Gas Pressure Differential Switch High for Air Heater (AHTR-21)	Pressure Differential Switch High	Internal wiring
Gas Flame Detector for Air Heater (AHTR-21)	Flame Detector	Internal wiring
Gas Flame Detector for Air Heater (AHTR-21)	Flame Detector	Internal wiring
Gas Ignition Transformer for Air Heater (AHTR-21)	Ignition Transformer	Internal wiring
Gas Flow Control Valve for Air Heater Gas Train (GTRN-21)	Flow Control Valve	Internal wiring
Gas Position Switch Low for Air Heater Gas Train (GTRN-21)	Position Switch Low	Internal wiring
Gas Position Transmitter for Air Heater Gas Train (GTRN-21)	Position Transmitter	Internal wiring
Gas Pressure Switch High for Air Heater Gas Train (GTRN-21)	Pressure Switch High	Internal wiring
Gas Blocking Valve (main gas) for Air Heater Gas Train (GTRN-21)	Blocking Valve (main gas)	Internal wiring
Gas Position Switch Open for Air Heater Gas Train (GTRN-21)	Position Switch Open	Internal wiring
Gas Position Switch Close for Air Heater Gas Train (GTRN-21)	Position Switch Close	Internal wiring
Gas Venting Valve (main gas) for Air Heater Gas Train (GTRN-21)	Venting Valve (main gas)	Internal wiring
Gas Position Switch Open for Air Heater Gas Train (GTRN-21)	Position Switch Open	Internal wiring
Gas Position Switch Close for Air Heater Gas Train (GTRN-21)	Position Switch Close	Internal wiring
Gas Pressure Switch Low for Air Heater Gas Train (GTRN-21)	Pressure Switch Low	Internal wiring
Gas Blocking Valve (main gas) for Air Heater Gas Train (GTRN-21)	Blocking Valve (main gas)	Internal wiring
Gas Position Switch Open for Air Heater Gas Train (GTRN-21)	Position Switch Open	Internal wiring
Gas Position Switch Close for Air Heater Gas Train (GTRN-21)	Position Switch Close	Internal wiring
		-
E-Stop for PUPD-11 (Positive Displacement Pump)	E-Stop	Dry Contact
High Oil Pressure Alarm	Lamp	24 VDC energized
Feed Slurry Pressure Transmitter for High Pressure Pump Outlet (PUPD-11)	Pressure Transmitter	4 - 20 mA
Cooling Water Return Flow Switch for Positive Displacement Pump (PUPD-11)	Flow Switch	-
Start / Stop Command for PUPD-11 (Positive Displacement Pump)	Start / Stop Command	-
Vibration Gear Box Vibration Switch for Switch Vibration for PUPD-11 (PUPD-11)	Vibration Switch	4 - 20 mA
Cooling Water On/Off Valve for Cooling Water Supply for PUPD-11 (PUPD-11)	On/Off Valve	24 VDC energized
Oil Pressure Switch for Positive Displacement Pump (PUPD-11)	Flow Switch	-
Temperature for Positive Displacement Pump (PUPD-11)	TC / Temperature Transmitter	-
Temperature for Positive Displacement Pump (PUPD-11)	TC / Temperature Transmitter	-
Fault Command		-
Run Command		-
		-

Gas and Powder Pressure Switch High for Storage Bin Cyclone Inlet Duct (####-##)	Pressure Switch High	-
Gas Pressure Switch High for Spray Chamber (CHAM-21)	Pressure Switch High	-
Gas Pressure Switch High for Cyclone (CYCL-21)	Pressure Switch High	-
Gas Pressure Switch High for Cyclone (CYCL-22)	Pressure Switch High	-
Gas and Powder Pressure Switch High for Bag House (BGHS-21)	Pressure Switch High	-
General Alarm Emergency Stop Actuation for XSUP-5601 (Supression Panel)	General Alarm Emergency Stop Actuation	Dry Contact
General Fault Controlled Stop for XSUP-5601 (Supression Panel)	General Fault Controlled Stop	Dry Contact
General Armed Status Indicator for XSUP-5601 (Supression Panel)	General Armed Status Indicator	Dry Contact
Power Fault Status Indicator for XSUP-5601 (Supression Panel)	Power Fault Status Indicator	Dry Contact
Dryer CHAM-21 Emergency Stop Actuation for XSUP-5601 (Supression Panel)	Dryer CHAM-21 Emergency Stop Actuation	Dry Contact
CHAM-21 Controlled Stop Trouble (No Protection) for XSUP-5601 (Supression Panel)	CHAM-21 Controlled Stop Trouble (No Protection)	Dry Contact
CYCL-21/22 Emergency Stop Actuation for XSUP-5601 (Supression Panel)	CYCL-21/22 Emergency Stop Actuation	Dry Contact
CYCL-21/22 Controlled Stop Trouble (No Protection) for XSUP-5601 (Supression Panel)	CYCL-21/22 Controlled Stop Trouble (No Protection)	Dry Contact
BGHS-21 Emergency Stop Actuation for XSUP-5601 (Supression Panel)	BGHS-21 Emergency Stop Actuation	Dry Contact
BGHS-21 Controlled Stop Trouble (No Protection) for XSUP-5601 (Supression Panel)	BGHS-21 Controlled Stop Trouble (No Protection)	Dry Contact
CYCL-31 Emergency Stop Actuation for XSUP-5601 (Supression Panel)	CYCL-31 Emergency Stop Actuation	Dry Contact
CYCL-31 Controlled Stop Trouble (No Protection) for XSUP-5601 (Supression Panel)	CYCL-31 Controlled Stop Trouble (No Protection)	Dry Contact
		-
Fire Deluge Fault (No Protection) for FWDP-21 (Fire Deluge)	Fire Deluge Fault (No Protection)	Dry Contact
Fire Deluge Alarm (Deluge Activated) for FWDP-21 (Fire Deluge)	Fire Deluge Alarm (Deluge Activated)	Dry Contact
Start / Stop Command for FWDP-21 (Fire Deluge)	Start / Stop Command	24 VDC energized
Gas and Powder Temperature Switch High for Conveying Duct (DTCO-21)	Temperature Switch	Dry Contact
		-
Fire Deluge Fault (No Protection) for FWDP-21 (Fire Deluge)	Fire Deluge Fault (No Protection)	Dry Contact
Fire Deluge Alarm (Deluge Activated) for FWDP-21 (Fire Deluge)	Fire Deluge Alarm (Deluge Activated)	Dry Contact
Start / Stop Command for FWDP-21 (Fire Deluge)	Start / Stop Command	24 VDC energized
Drain Valve		24 VDC energized
Gas and Powder Temperature Switch High for Exhaust Duct (DTEX-22)	Temperature Switch	Dry Contact
		-
Start / Stop Command for Bin Aerator Blower	Start / Stop Command	Dry Contact
Operational / Running status		Dry Contact
Local / Remote status		Dry Contact
Fault / Trip condition		Dry Contact
Warning condition		Dry Contact
		-

Slurry Drain line	Valve With Position Switch	Dry Contact
Slurry Drain line	Valve With Position Switch	Dry Contact
Slurry Drain line	Valve With Position Switch	Dry Contact
Slurry Drain line	Valve With Position Switch	Dry Contact
Slurry Drain line	Valve With Position Switch	Dry Contact
Slurry Drain line	Valve With Position Switch	Dry Contact
Slurry Drain line	Valve With Position Switch	Dry Contact
Slurry Preparation Tank 1	Automatic on/off valve	24 VDC energized
Slurry Preparation Tank 1	Automatic on/off valve	24 VDC energized
Slurry Preparation Tank 1	Automatic on/off valve	24 VDC energized
Slurry Preparation Tank 2	Automatic on/off valve	24 VDC energized
Slurry Preparation Tank 2	Automatic on/off valve	24 VDC energized
Slurry Preparation Tank 2	Automatic on/off valve	24 VDC energized
Level alarm low	Level switch	Dry Contact
Start / Stop Command for FVNR PC-2030		24 VDC energized
Start / Stop Command for FVNR PC-2031		24 VDC energized
Start / Stop Command to VSD PC-81001		24 VDC energized
Start / Stop Command to VSD PC-81002		24 VDC energized
		-
Flow outlet LPG line	Coriolis Flow Transmitter	4 - 20 mA,Pulse
Level LPG in storage	Level transmitter	4 - 20 mA
Pressure outlet LPG line	Pressure Transmitter	4 - 20 mA
Outlet LPG line	Automatic on/off valve	24 VDC energized
		-
Gas Vaporizer 4 Run Status	Gas vaporizer Ready	Dry Contact
Gas Vaporizer 4 Alarm status	Gas vaporizer Run	Dry Contact
Gas Vaporizer 4 High Temp Shut-Down status	Gas vaporizer Failed	Dry Contact
Gas Vaporizer 4 Working Temp Shun-Off status	Gas vaporizer Overheated	Dry Contact
Gas Vaporizer 4 Low Water Shun-Off status	Gas vaporizer Excessive flow tripped	Dry Contact
Start Command for Gas Vaporizer 4	Start Command	24 VDC energized
Stop Command for Gas Vaporizer 4	Stop Command	24 VDC energized
		-
Gas Vaporizer 5 Run Status	Gas vaporizer Ready	Dry Contact
Gas Vaporizer 5 Alarm status	Gas vaporizer Run	Dry Contact
Gas Vaporizer 5 High Temp Shut-Down status	Gas vaporizer Failed	Dry Contact

Gas Vaporizer 5 Working Temp Shun-Off status	Gas vaporizer Overheated	Dry Contact
Gas Vaporizer 5 Low Water Shun-Off status	Gas vaporizer Excessive flow tripped	Dry Contact
Start Command for Gas Vaporizer 5	Start Command	24 VDC energized
Stop Command for Gas Vaporizer 5	Stop Command	24 VDC energized

KALASIN ENGINEERING | MASTER INSTRUMENT

Point-by-Point Field Loop Check Record (Rev. 3.6a M

[illegible]

0 to 100 l/h	TYP-AI-02	TBAI	LOOP DIAGRAM WRING.dwg	JB-607
0 to 100 l/h	TYP-AI-02	TBAI	LOOP DIAGRAM WRING.dwg	JB-607
0 to 100 l/h	TYP-AI-02	TBAI	LOOP DIAGRAM WRING.dwg	JB-607
0 to 100 l/h	TYP-AI-02	TBAI	LOOP DIAGRAM WRING.dwg	JB-607
0 to 100 l/h	TYP-AI-02	TBAI	LOOP DIAGRAM WRING.dwg	JB-607
0 to 100 l/h	TYP-AI-02	TBAI	LOOP DIAGRAM WRING.dwg	JB-607
0 - 1000 m3/hr	TYP-AI-02	TBAI	LOOP DIAGRAM WRING.dwg	JB-601
0 - 7000 kg/hr	TYP-AI-02	TBAI	LOOP DIAGRAM WRING.dwg	JB-606
0 - 2000 kg/hr	TYP-AI-02	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
0 - 1000 m3/hr	TYP-AI-02	TBAI	LOOP DIAGRAM WRING.dwg	JB-601
0-1000kg/m ³	TYP-AI-02	TBAI	LOOP DIAGRAM WRING.dwg	RIO-200
0 - 20 m3/ hr	TYP-AI-02	TBAI	LOOP DIAGRAM WRING.dwg	RIO-200
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-401
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-401
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-401
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-401
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-401

On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-401
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-402
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-402
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-603
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-603
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-603
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-606
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-602
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-601
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-618
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-618
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-618
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-612
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-612
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-618
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-612
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-618
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-612
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
0 - 10 m.	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-401
0 - 10 m.	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-401
0 - 10 m.	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-401
-	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-618
0-14	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-401
0-14	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-401
-0.01 mbar	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-602
0.3-0.7barg	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-618
40 mbarg	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-618
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-612
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-612
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-618
-100 to 500 mBar	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-401
0 - 10'barg	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-401
0 - 12'barg	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-402
0 - 20 bar	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-607

0 to 500 bar	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-401
0 - 0.1 mbar	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-601
0 - 0.1 mbar	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
0 - 6 bar	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
0 - 100 mbar	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
-100 - 100 mbar	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-608
0 - 0.1 mbar	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-607
0 - 0.1 mbar	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-603
-100 to 200 mbar	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-603
-100 to 200 mbar	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
0 - 0.1 mbar	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
0 - 10 barg	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-601
0 - 1 barg	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-618
-100 - 100 mbarg	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-612
-100 to 200 mbar	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-618
-100 to 200 mbar	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-618
-100 to 200 mbar	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-618
0 - 10 barg	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-612
0 - 100%	TYP-AO-01	TBAO	LOOP DIAGRAM WRING.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-401
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-401
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-401
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-401
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-606
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-601
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-601
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-601
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-601
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-601
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-618
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-618
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-618
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-618
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-612

On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-612
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-612
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-612
0 - 140 C	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-602
0 - 140 C	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-602
0 - 120 C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-402
0 - 200 C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-402
0 - 300 C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-607
0 - 400°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
0 - 100 C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
0 - 100 C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
0 - 400°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
-10 to 250°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
-10 to 150°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
-10 to 150°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
0 - 200°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
0 - 400°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-608
0 - 400°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-608
0 - 400°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-608
0 - 400°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-608
0 - 400°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-603
0 - 400°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-603
0 - 400°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-603
0 - 400°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-603
0 - 400°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-603
0 - 400°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
0 - 100 C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
0 - 100 C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-602
-10 to 150°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-608
0 - 40 C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-607
0 - 400°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-618
0 - 400°C	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	IS-JB-618
0 - 100%	TYP-AO-01	TBAO	LOOP DIAGRAM WRING.dwg	JB-402
0 - 100%	TYP-AO-01	TBAO	LOOP DIAGRAM WRING.dwg	JB-602
0 - 100%	TYP-AO-01	TBAO	LOOP DIAGRAM WRING.dwg	PLC

[illegible]

[illegible]

[illegible]

On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-607
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-602
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-618
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-618
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC

On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-612
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-612
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-618
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-618
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	IS-JB-618
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	IS-JB-612
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	IS-JB-612
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	IS-JB-603
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	IS-JB-603
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	IS-JB-603
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	JB-612
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC

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On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	PLC
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	PLC
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	PLC
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	PLC
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-602
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-602
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-602
-	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-601
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-602
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-602
0 - 100 C	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-602
0 - 100 C	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-602
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-602
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-602
0 - 100 C	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-602
0 - 100 C	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-602

-	TYP-MECH	N/A	<i>Standard Mechanical Typical</i>	Local Dial
On-Off	TYP-DO-01	TBDO	<i>PANEL WIRING DIAGRAM.dwg</i>	PLC
On-Off	TYP-DI-01	TBDI	<i>PANEL WIRING DIAGRAM.dwg</i>	JB-602
On-Off	TYP-DI-01	TBDI	<i>PANEL WIRING DIAGRAM.dwg</i>	JB-602
0 - 100 C	TYP-DI-01	TBDI	<i>PANEL WIRING DIAGRAM.dwg</i>	JB-602
0 - 100 C	TYP-DI-01	TBDI	<i>PANEL WIRING DIAGRAM.dwg</i>	JB-602
-	TYP-MECH	N/A	<i>Standard Mechanical Typical</i>	Local Dial
On-Off	TYP-DI-01	TBDI	<i>PANEL WIRING DIAGRAM.dwg</i>	JB-607
On-Off	TYP-DI-01	TBDI	<i>PANEL WIRING DIAGRAM.dwg</i>	JB-607
On-Off	TYP-DI-01	TBDI	<i>PANEL WIRING DIAGRAM.dwg</i>	JB-607
On-Off	TYP-DI-01	TBDI	<i>PANEL WIRING DIAGRAM.dwg</i>	JB-607
On-Off	TYP-DI-01	TBDI	<i>PANEL WIRING DIAGRAM.dwg</i>	JB-607
On-Off	TYP-DO-01	TBDO	<i>PANEL WIRING DIAGRAM.dwg</i>	PLC
On-Off	TYP-DO-01	TBDO	<i>PANEL WIRING DIAGRAM.dwg</i>	PLC
On-Off	TYP-DO-01	TBDO	<i>PANEL WIRING DIAGRAM.dwg</i>	JB-607
On-Off	TYP-DO-01	TBDO	<i>PANEL WIRING DIAGRAM.dwg</i>	JB-607
On-Off	TYP-DO-01	TBDO	<i>PANEL WIRING DIAGRAM.dwg</i>	PLC
On-Off	TYP-DO-01	TBDO	<i>PANEL WIRING DIAGRAM.dwg</i>	PLC
On-Off	TYP-DO-01	TBDO	<i>PANEL WIRING DIAGRAM.dwg</i>	PLC
On-Off	TYP-AI-01	TBAI	<i>LOOP DIAGRAM WRING.dwg</i>	JB-607
On-Off	TYP-AI-01	TBAI	<i>LOOP DIAGRAM WRING.dwg</i>	JB-607
On-Off	TYP-AO-01	TBAO	<i>LOOP DIAGRAM WRING.dwg</i>	JB-607
-	TYP-GEN-01	TBCI	<i>LOOP DIAGRAM WRING.dwg</i>	PLC
-	TYP-GEN-01	TBCI	<i>LOOP DIAGRAM WRING.dwg</i>	AHTR Control Panel(HCP)
-	TYP-GEN-01	TBCI	<i>LOOP DIAGRAM WRING.dwg</i>	AHTR Control Panel(HCP)
-	TYP-DI-01	TBDI	<i>PANEL WIRING DIAGRAM.dwg</i>	AHTR Control Panel(HCP)
-	TYP-AO-01	TBAO	<i>LOOP DIAGRAM WRING.dwg</i>	AHTR Control Panel(HCP)
-	TYP-GEN-01	TBCI	<i>LOOP DIAGRAM WRING.dwg</i>	AHTR Control Panel(HCP)
-	TYP-GEN-01	TBCI	<i>LOOP DIAGRAM WRING.dwg</i>	AHTR Control Panel(HCP)
-	TYP-GEN-01	TBCI	<i>LOOP DIAGRAM WRING.dwg</i>	AHTR Control Panel(HCP)
-	TYP-GEN-01	TBCI	<i>LOOP DIAGRAM WRING.dwg</i>	AHTR Control Panel(HCP)
-	TYP-GEN-01	TBCI	<i>LOOP DIAGRAM WRING.dwg</i>	AHTR Control Panel(HCP)
-	TYP-GEN-01	TBCI	<i>LOOP DIAGRAM WRING.dwg</i>	AHTR Control Panel(HCP)
-	TYP-GEN-01	TBCI	<i>LOOP DIAGRAM WRING.dwg</i>	AHTR Control Panel(HCP)
-	TYP-GEN-01	TBCI	<i>LOOP DIAGRAM WRING.dwg</i>	AHTR Control Panel(HCP)
-	TYP-AO-01	TBAO	<i>LOOP DIAGRAM WRING.dwg</i>	AHTR Control Panel

-	TYP-AO-01	TBAO	LOOP DIAGRAM WRING.dwg	AHTR Control Panel
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	AHTR Control Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	AHTR Control Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	AHTR Control Panel
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	AHTR Control Panel
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	AHTR Control Panel
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	AHTR Control Panel
-	TYP-AO-01	TBAO	LOOP DIAGRAM WRING.dwg	AHTR Control Panel
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	AHTR Control Panel
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	AHTR Control Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	AHTR Control Panel
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	AHTR Control Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	AHTR Control Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	AHTR Control Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	AHTR Control Panel
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	AHTR Control Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	AHTR Control Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	AHTR Control Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	AHTR Control Panel
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	AHTR Control Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	AHTR Control Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	AHTR Control Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	AHTR Control Panel
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	AHTR Control Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	AHTR Control Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	AHTR Control Panel
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-601
-	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
-	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-401
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-601
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	JB-601
-	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-601
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	PLC
-	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	PLC
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	JB-601
-	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	JB-601
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial

-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	Suppression Sys. Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	Suppression Sys. Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	Suppression Sys. Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	Suppression Sys. Panel
-	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	Suppression Sys. Panel
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-601
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-601
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-601
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-601
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-601
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-601
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-601
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-601
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-601
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-601
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-601
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-601
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-606
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-606
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
0 - 140 C	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-602
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	JB-601
0 - 140 C	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	JB-602
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	PLC
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	PLC
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial

On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	RIO-200
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	RIO-200
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	RIO-200
On-Off	TYP-GEN-01	TBCI	LOOP DIAGRAM WRING.dwg	RIO-200
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial
0-1000kg/hr	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
-	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	RIO-200
0 - 10 barg	TYP-AI-01	TBAI	LOOP DIAGRAM WRING.dwg	RIO-200
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	RIO-200
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	RIO-200
-	TYP-MECH	N/A	Standard Mechanical Typical	Local Dial
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200

On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DI-01	TBDI	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	RIO-200
On-Off	TYP-DO-01	TBDO	PANEL WIRING DIAGRAM.dwg	RIO-200

PLC LOOP TESTING & CALIBRATION VERIFICATION SCHEDULE

(aster) Cross-Referenced with CAD Loop Typical Diagrams (DWG/DXF)

Terminal Block ID	PLC Rack / Slot	Wire Tag (PLC Side)	Wire Tag (Terminal Side)
P2-TBAI2-1(B)	C2/C2S10	C2S10-1/P2-TBAI2-1(B)	P2-TBAI2-1(B)/C2S10-1
P3-TBAI1-15(B)	C3/C3S6	C3S6-33/P3-TBAI1-15(B)	P3-TBAI1-15(B)/C3S6-33
P4-TBAI2-5(B)	C4/C4S6	C4S6-11/P4-TBAI2-5(B)	P4-TBAI2-5(B)/C4S6-11
P4-TBAI2-6(B)	C4/C4S6	C4S6-13/P4-TBAI2-6(B)	P4-TBAI2-6(B)/C4S6-13
P4-TBAI2-1(B)	C4/C4S6	C4S6-1/P4-TBAI2-1(B)	P4-TBAI2-1(B)/C4S6-1
P4-TBAI2-2(B)	C4/C4S6	C4S6-3/P4-TBAI2-2(B)	P4-TBAI2-2(B)/C4S6-3
P1-TBAI2-2(B)	C1/C1S11	C1S11-3/P1-TBAI2-2(B)	P1-TBAI2-2(B)/C1S11-3
P2-TBAI2-2(B)	C2/C2S10	C2S10-3/P2-TBAI2-2(B)	P2-TBAI2-2(B)/C2S10-3
-	-	-	-
P1-TBDI1-17	C1/C1S4	C1S4-19/P1-TBDI1-17	P1-TBDI1-17/C1S4-19
P1-TBDI1-18	C1/C1S4	C1S4-20/P1-TBDI1-18	P1-TBDI1-18/C1S4-20
P2-TBDI1-1	C2/C2S1	C2S1-1/P2-TBDI1-1	P2-TBDI1-1/C2S1-1
P2-TBDI1-2	C2/C2S1	C2S1-2/P2-TBDI1-2	P2-TBDI1-2/C2S1-2
P1-TBDI1-19	C1/C1S4	C1S4-21/P1-TBDI1-19	P1-TBDI1-19/C1S4-21
P1-TBDI1-20	C1/C1S4	C1S4-22/P1-TBDI1-20	P1-TBDI1-20/C1S4-22
P2-TBDI1-3	C2/C2S1	C2S1-3/P2-TBDI1-3	P2-TBDI1-3/C2S1-3
P2-TBDI1-4	C2/C2S1	C2S1-4/P2-TBDI1-4	P2-TBDI1-4/C2S1-4
P3-TBAI2-2(B)	C3/C3S7	C3S7-3/P3-TBAI2-2(B)	P3-TBAI2-2(B)/C3S7-3
P1-TBAI2-3(B)	C1/C1S11	C1S11-5/P1-TBAI2-3(B)	P1-TBAI2-3(B)/C1S11-5
P3-TBAI1-10(B)	C3/C3S6	C3S6-21/P3-TBAI1-10(B)	P3-TBAI1-10(B)/C3S6-21
P3-TBAI1-11(B)	C3/C3S6	C3S6-23/P3-TBAI1-11(B)	P3-TBAI1-11(B)/C3S6-23
P3-TBAI2-3(B)	C3/C3S7	C3S7-5/P3-TBAI2-3(B)	P3-TBAI2-3(B)/C3S7-5
P3-TBAI2-4(B)	C3/C3S7	C3S7-7/P3-TBAI2-4(B)	P3-TBAI2-4(B)/C3S7-7
P3-TBAI2-5(B)	C3/C3S7	C3S7-11/P3-TBAI2-5(B)	P3-TBAI2-5(B)/C3S7-11
P3-TBAI2-6(B)	C3/C3S7	C3S7-13/P3-TBAI2-6(B)	P3-TBAI2-6(B)/C3S7-13
P3-TBAI2-7(B)	C3/C3S7	C3S7-15/P3-TBAI2-7(B)	P3-TBAI2-7(B)/C3S7-15
P3-TBAI2-8(B)	C3/C3S7	C3S7-17/P3-TBAI2-8(B)	P3-TBAI2-8(B)/C3S7-17
P3-TBAI2-9(B)	C3/C3S7	C3S7-19/P3-TBAI2-9(B)	P3-TBAI2-9(B)/C3S7-19
P3-TBAI2-10(B)	C3/C3S7	C3S7-21/P3-TBAI2-10(B)	P3-TBAI2-10(B)/C3S7-21
P3-TBAI2-11(B)	C3/C3S7	C3S7-23/P3-TBAI2-11(B)	P3-TBAI2-11(B)/C3S7-23
P3-TBAI2-12(B)	C3/C3S7	C3S7-25/P3-TBAI2-12(B)	P3-TBAI2-12(B)/C3S7-25

P3-TBAI2-13(B)	C3/C3S7	C3S7-29/P3-TBAI2-13(B)	P3-TBAI2-13(B)/C3S7-29
P3-TBAI2-14(B)	C3/C3S7	C3S7-31/P3-TBAI2-14(B)	P3-TBAI2-14(B)/C3S7-31
P3-TBAI2-15(B)	C3/C3S7	C3S7-33/P3-TBAI2-15(B)	P3-TBAI2-15(B)/C3S7-33
P3-TBAI2-16(B)	C3/C3S7	C3S7-35/P3-TBAI2-16(B)	P3-TBAI2-16(B)/C3S7-35
P3-TBAI3-1(B)	C3/C3S8	C3S8-1/P3-TBAI3-1(B)	P3-TBAI3-1(B)/C3S8-1
P3-TBAI3-2(B)	C3/C3S8	C3S8-3/P3-TBAI3-2(B)	P3-TBAI3-2(B)/C3S8-3
P1-TBAI2-5(B)	C1/C1S11	C1S11-11/P1-TBAI2-5(B)	P1-TBAI2-5(B)/C1S11-11
P3-TBDI1-17	C3/C3S1	C3S1-19/P3-TBDI1-17	P3-TBDI1-17/C3S1-19
P2-TBDI2-1	C2/C2S2	C2S2-1/P2-TBDI2-1	P2-TBDI2-1/C2S2-1
P1-TBAI2-4(B)	C1/C1S11	C1S11-7/P1-TBAI2-4(B)	P1-TBAI2-4(B)/C1S11-7
P5-TBDI1-3	C5/C5S1	C5S1-3/P5-TBDI1-3	P5-TBDI1-3/C5S1-3
P5-TBDI1-4	C5/C5S1	C5S1-4/P5-TBDI1-4	P5-TBDI1-4/C5S1-4
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
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-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
P1-TBDI1-23	C1/C1S4	C1S4-25/P1-TBDI1-23	P1-TBDI1-23/C1S4-25
P1-TBDI1-21	C1/C1S4	C1S4-23/P1-TBDI1-21	P1-TBDI1-21/C1S4-23
P1-TBDI1-24	C1/C1S4	C1S4-26/P1-TBDI1-24	P1-TBDI1-24/C1S4-26
P1-TBDI1-22	C1/C1S4	C1S4-24/P1-TBDI1-22	P1-TBDI1-22/C1S4-24
P1-TBDI1-26	C1/C1S4	C1S4-28/P1-TBDI1-26	P1-TBDI1-26/C1S4-28

P1-TBDI1-25	C1/C1S4	C1S4-27/P1-TBDI1-25	P1-TBDI1-25/C1S4-27
P2-TBDI1-5	C2/C2S1	C2S1-5/P2-TBDI1-5	P2-TBDI1-5/C2S1-5
P2-TBDI1-6	C2/C2S1	C2S1-6/P2-TBDI1-6	P2-TBDI1-6/C2S1-6
P4-TBDI1-4	C4/C4S1	C4S1-4/P4-TBDI1-4	P4-TBDI1-4/C4S1-4
P4-TBDI1-5	C4/C4S1	C4S1-5/P4-TBDI1-5	P4-TBDI1-5/C4S1-5
P4-TBDI1-6	C4/C4S1	C4S1-6/P4-TBDI1-6	P4-TBDI1-6/C4S1-6
P3-TBDI1-18	C3/C3S1	C3S1-20/P3-TBDI1-18	P3-TBDI1-18/C3S1-20
P2-TBDI2-2	C2/C2S2	C2S2-2/P2-TBDI2-2	P2-TBDI2-2/C2S2-2
P1-TBDI3-19	C1/C1S6	C1S6-21/P1-TBDI3-19	P1-TBDI3-19/C1S6-21
P4-TBDI3-2	C4/C4S3	C4S3-2/P4-TBDI3-2	P4-TBDI3-2/C4S3-2
P4-TBDI3-6	C4/C4S3	C4S3-6/P4-TBDI3-6	P4-TBDI3-6/C4S3-6
P3-TBDI3-5	C3/C3S3	C3S3-5/P3-TBDI3-5	P3-TBDI3-5/C3S3-5
P4-TBDI2-3	C4/C4S2	C4S2-3/P4-TBDI2-3	P4-TBDI2-3/C4S2-3
P4-TBDI2-4	C4/C4S2	C4S2-4/P4-TBDI2-4	P4-TBDI2-4/C4S2-4
P4-TBDI3-4	C4/C4S3	C4S3-4/P4-TBDI3-4	P4-TBDI3-4/C4S3-4
P4-TBDI2-5	C4/C4S2	C4S2-5/P4-TBDI2-5	P4-TBDI2-5/C4S2-5
P4-TBDI3-5	C4/C4S3	C4S3-5/P4-TBDI3-5	P4-TBDI3-5/C4S3-5
P4-TBDI2-6	C4/C4S2	C4S2-6/P4-TBDI2-6	P4-TBDI2-6/C4S2-6
P5-TBDI1-5	C5/C5S1	C5S1-5/P5-TBDI1-5	P5-TBDI1-5/C5S1-5
P1-TBAI1-5(B)	C1/C1S10	C1S10-11/P1-TBAI1-5(B)	P1-TBAI1-5(B)/C1S10-11
P1-TBAI1-6(B)	C1/C1S10	C1S10-13/P1-TBAI1-6(B)	P1-TBAI1-6(B)/C1S10-13
P1-TBAI1-4(B)	C1/C1S10	C1S10-7/P1-TBAI1-4(B)	P1-TBAI1-4(B)/C1S10-7
P4-TBAI2-7(B)	C4/C4S6	C4S6-15/P4-TBAI2-7(B)	P4-TBAI2-7(B)/C4S6-15
P1-TBAI1-7(B)	C1/C1S10	C1S10-15/P1-TBAI1-7(B)	P1-TBAI1-7(B)/C1S10-15
P1-TBAI1-8(B)	C1/C1S10	C1S10-17/P1-TBAI1-8(B)	P1-TBAI1-8(B)/C1S10-17
P2-TBDI2-3	C2/C2S2	C2S2-3/P2-TBDI2-3	P2-TBDI2-3/C2S2-3
P3-TBDI3-6	C3/C3S3	C3S3-6/P3-TBDI3-6	P3-TBDI3-6/C3S3-6
P3-TBDI3-7	C3/C3S3	C3S3-7/P3-TBDI3-7	P3-TBDI3-7/C3S3-7
P3-TBDI2-1	C3/C3S2	C3S2-1/P3-TBDI2-1	P3-TBDI2-1/C3S2-1
P3-TBDI2-2	C3/C3S2	C3S2-2/P3-TBDI2-2	P3-TBDI2-2/C3S2-2
P3-TBDI3-8	C3/C3S3	C3S3-8/P3-TBDI3-8	P3-TBDI3-8/C3S3-8
P1-TBAI1-9(B)	C1/C1S10	C1S10-19/P1-TBAI1-9(B)	P1-TBAI1-9(B)/C1S10-19
P1-TBAI1-10(B)	C1/C1S10	C1S10-21/P1-TBAI1-10(B)	P1-TBAI1-10(B)/C1S10-21
P2-TBAI1-5(B)	C2/C2S9	C2S9-11/P2-TBAI1-5(B)	P2-TBAI1-5(B)/C2S9-11
P3-TBAI3-3(B)	C3/C3S8	C3S8-5/P3-TBAI3-3(B)	P3-TBAI3-3(B)/C3S8-5

P1-TBAI1-11(B)	C1/C1S10	C1S10-23/P1-TBAI1-11(B)	P1-TBAI1-11(B)/C1S10-23
P1-TBAI2-6(B)	C1/C1S11	C1S11-13/P1-TBAI2-6(B)	P1-TBAI2-6(B)/C1S11-13
P2-TBAI2-4(B)	C2/C2S10	C2S10-7/P2-TBAI2-4(B)	P2-TBAI2-4(B)/C2S10-7
P2-TBAI2-5(B)	C2/C2S10	C2S10-11/P2-TBAI2-5(B)	P2-TBAI2-5(B)/C2S10-11
P2-TBAI2-6(B)	C2/C2S10	C2S10-13/P2-TBAI2-6(B)	P2-TBAI2-6(B)/C2S10-13
P4-TBAI1-9(B)	C4/C4S5	C4S5-19/P4-TBAI1-9(B)	P4-TBAI1-9(B)/C4S5-19
P3-TBAI3-5(B)	C3/C3S8	C3S8-11/P3-TBAI3-5(B)	P3-TBAI3-5(B)/C3S8-11
P4-TBAI1-1(B)	C4/C4S5	C4S5-1/P4-TBAI1-1(B)	P4-TBAI1-1(B)/C4S5-1
P4-TBAI1-2(B)	C4/C4S5	C4S5-3/P4-TBAI1-2(B)	P4-TBAI1-2(B)/C4S5-3
P2-TBAI2-7(B)	C2/C2S10	C2S10-15/P2-TBAI2-7(B)	P2-TBAI2-7(B)/C2S10-15
P2-TBAI2-8(B)	C2/C2S10	C2S10-17/P2-TBAI2-8(B)	P2-TBAI2-8(B)/C2S10-17
P1-TBAI2-7(B)	C1/C1S11	C1S11-15/P1-TBAI2-7(B)	P1-TBAI2-7(B)/C1S11-15
P4-TBAI2-8(B)	C4/C4S6	C4S6-17/P4-TBAI2-8(B)	P4-TBAI2-8(B)/C4S6-17
P3-TBAI4-9(B)	C3/C3S9	C3S9-19/P3-TBAI4-9(B)	P3-TBAI4-9(B)/C3S9-19
P3-TBAI4-1(B)	C3/C3S9	C3S9-1/P3-TBAI4-1(B)	P3-TBAI4-1(B)/C3S9-1
P4-TBAI2-10(B)	C4/C4S6	C4S6-21/P4-TBAI2-10(B)	P4-TBAI2-10(B)/C4S6-21
P4-TBAI2-11(B)	C4/C4S6	C4S6-23/P4-TBAI2-11(B)	P4-TBAI2-11(B)/C4S6-23
P3-TBAI4-10(B)	C3/C3S9	C3S9-21/P3-TBAI4-10(B)	P3-TBAI4-10(B)/C3S9-21
P3-TBAO1-4	C3/C3S12	C3S12-19/P3-TBAO1-4	P3-TBAO1-4/C3S12-19
P1-TBRL1-25	C1/C1S8	C1S8-27/P1-TBRL1-25	P1-TBRL1-25/C1S8-27
P1-TBRL1-26	C1/C1S8	C1S8-28/P1-TBRL1-26	P1-TBRL1-26/C1S8-28
P1-TBRL1-27	C1/C1S8	C1S8-29/P1-TBRL1-27	P1-TBRL1-27/C1S8-29
P1-TBRL1-28	C1/C1S8	C1S8-30/P1-TBRL1-28	P1-TBRL1-28/C1S8-30
P3-TBRL1-9	C3/C3S4	C3S4-9/P3-TBRL1-9	P3-TBRL1-9/C3S4-9
P1-TBRL2-19	C1/C1S9	C1S9-21/P1-TBRL2-19	P1-TBRL2-19/C1S9-21
P1-TBRL2-22	C1/C1S9	C1S9-24/P1-TBRL2-22	P1-TBRL2-22/C1S9-24
P1-TBRL2-21	C1/C1S9	C1S9-23/P1-TBRL2-21	P1-TBRL2-21/C1S9-23
P1-TBRL2-23	C1/C1S9	C1S9-25/P1-TBRL2-23	P1-TBRL2-23/C1S9-25
P1-TBRL2-24	C1/C1S9	C1S9-26/P1-TBRL2-24	P1-TBRL2-24/C1S9-26
P3-TBRL2-3	C3/C3S5	C3S5-3/P3-TBRL2-3	P3-TBRL2-3/C3S5-3
P3-TBRL2-4	C3/C3S5	C3S5-4/P3-TBRL2-4	P3-TBRL2-4/C3S5-4
P3-TBRL2-8	C3/C3S5	C3S5-8/P3-TBRL2-8	P3-TBRL2-8/C3S5-8
P3-TBRL2-5	C3/C3S5	C3S5-5/P3-TBRL2-5	P3-TBRL2-5/C3S5-5
P3-TBRL2-7	C3/C3S5	C3S5-7/P3-TBRL2-7	P3-TBRL2-7/C3S5-7
P3-TBRL1-20	C3/C3S4	C3S4-22/P3-TBRL1-20	P3-TBRL1-20/C3S4-22

P3-TBRL1-21	C3/C3S4	C3S4-23/P3-TBRL1-21	P3-TBRL1-21/C3S4-23
P3-TBRL1-22	C3/C3S4	C3S4-24/P3-TBRL1-22	P3-TBRL1-22/C3S4-24
P3-TBRL1-23	C3/C3S4	C3S4-25/P3-TBRL1-23	P3-TBRL1-23/C3S4-25
P2-TBDI2-4	C2/C2S2	C2S2-4/P2-TBDI2-4	P2-TBDI2-4/C2S2-4
P2-TBDI2-5	C2/C2S2	C2S2-5/P2-TBDI2-5	P2-TBDI2-5/C2S2-5
P2-TBAI1-6(B)	C2/C2S9	C2S9-13/P2-TBAI1-6(B)	P2-TBAI1-6(B)/C2S9-13
P2-TBAI1-7(B)	C2/C2S9	C2S9-15/P2-TBAI1-7(B)	P2-TBAI1-7(B)/C2S9-15
P3-TBAI3-8(B)	C3/C3S8	C3S8-17/P3-TBAI3-8(B)	P3-TBAI3-8(B)/C3S8-17
P2-TBAI2-9(B)	C2/C2S10	C2S10-19/P2-TBAI2-9(B)	P2-TBAI2-9(B)/C2S10-19
P2-TBAI2-10(B)	C2/C2S10	C2S10-21/P2-TBAI2-10(B)	P2-TBAI2-10(B)/C2S10-21
P2-TBAI2-11(B)	C2/C2S10	C2S10-23/P2-TBAI2-11(B)	P2-TBAI2-11(B)/C2S10-23
P2-TBAI2-12(B)	C2/C2S10	C2S10-25/P2-TBAI2-12(B)	P2-TBAI2-12(B)/C2S10-25
P2-TBAI2-15(B)	C2/C2S10	C2S10-33/P2-TBAI2-15(B)	P2-TBAI2-15(B)/C2S10-33
P2-TBAI2-16(B)	C2/C2S10	C2S10-35/P2-TBAI2-16(B)	P2-TBAI2-16(B)/C2S10-35
P3-TBAI1-1(B)	C3/C3S6	C3S6-1/P3-TBAI1-1(B)	P3-TBAI1-1(B)/C3S6-1
P3-TBAI1-2(B)	C3/C3S6	C3S6-3/P3-TBAI1-2(B)	P3-TBAI1-2(B)/C3S6-3
P4-TBAI1-10(B)	C4/C4S5	C4S5-21/P4-TBAI1-10(B)	P4-TBAI1-10(B)/C4S5-21
P4-TBAI1-11(B)	C4/C4S5	C4S5-23/P4-TBAI1-11(B)	P4-TBAI1-11(B)/C4S5-23
P4-TBAI1-12(B)	C4/C4S5	C4S5-25/P4-TBAI1-12(B)	P4-TBAI1-12(B)/C4S5-25
P4-TBAI1-13(B)	C4/C4S5	C4S5-29/P4-TBAI1-13(B)	P4-TBAI1-13(B)/C4S5-29
P4-TBAI1-3(B)	C4/C4S5	C4S5-5/P4-TBAI1-3(B)	P4-TBAI1-3(B)/C4S5-5
P4-TBAI1-4(B)	C4/C4S5	C4S5-7/P4-TBAI1-4(B)	P4-TBAI1-4(B)/C4S5-7
P4-TBAI1-5(B)	C4/C4S5	C4S5-11/P4-TBAI1-5(B)	P4-TBAI1-5(B)/C4S5-11
P4-TBAI1-6(B)	C4/C4S5	C4S5-13/P4-TBAI1-6(B)	P4-TBAI1-6(B)/C4S5-13
P4-TBAI1-7(B)	C4/C4S5	C4S5-15/P4-TBAI1-7(B)	P4-TBAI1-7(B)/C4S5-15
P3-TBAI1-3(B)	C3/C3S6	C3S6-5/P3-TBAI1-3(B)	P3-TBAI1-3(B)/C3S6-5
P3-TBAI1-4(B)	C3/C3S6	C3S6-7/P3-TBAI1-4(B)	P3-TBAI1-4(B)/C3S6-7
P3-TBAI1-5(B)	C3/C3S6	C3S6-11/P3-TBAI1-5(B)	P3-TBAI1-5(B)/C3S6-11
P3-TBAI3-16(B)	C3/C3S8	C3S8-35/P3-TBAI3-16(B)	P3-TBAI3-16(B)/C3S8-35
P3-TBAI3-9(B)	C3/C3S8	C3S8-19/P3-TBAI3-9(B)	P3-TBAI3-9(B)/C3S8-19
P4-TBAI2-12(B)	C4/C4S6	C4S6-25/P4-TBAI2-12(B)	P4-TBAI2-12(B)/C4S6-25
P4-TBAI2-13(B)	C4/C4S6	C4S6-29/P4-TBAI2-13(B)	P4-TBAI2-13(B)/C4S6-29
P3-TBAO1-1	C3/C3S12	C3S12-3/P3-TBAO1-1	P3-TBAO1-1/C3S12-3
P3-TBAO1-3	C3/C3S12	C3S12-13/P3-TBAO1-3	P3-TBAO1-3/C3S12-13
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P3-TBAI1-6(B)	C3/C3S6	C3S6-13/P3-TBAI1-6(B)	P3-TBAI1-6(B)/C3S6-13
P3-TBAI1-7(B)	C3/C3S6	C3S6-15/P3-TBAI1-7(B)	P3-TBAI1-7(B)/C3S6-15
P3-TBAI1-8(B)	C3/C3S6	C3S6-17/P3-TBAI1-8(B)	P3-TBAI1-8(B)/C3S6-17
P3-TBAI1-9(B)	C3/C3S6	C3S6-19/P3-TBAI1-9(B)	P3-TBAI1-9(B)/C3S6-19
P4-TBAI1-14(B)	C4/C4S5	C4S5-31/P4-TBAI1-14(B)	P4-TBAI1-14(B)/C4S5-31
P4-TBRL1-9	C4/C4S4	C4S4-9/P4-TBRL1-9	P4-TBRL1-9/C4S4-9
P5-TBAI1-8(B)	C5/C5S4	C5S4-17/P5-TBAI1-8(B)	P5-TBAI1-8(B)/C5S4-17
P5-TBRL1-1	C5/C5S3	C5S3-1/P5-TBRL1-1	P5-TBRL1-1/C5S3-1
P1-TBDI1-27	C1/C1S4	C1S4-29/P1-TBDI1-27	P1-TBDI1-27/C1S4-29
P1-TBDI1-29	C1/C1S4	C1S4-31/P1-TBDI1-29	P1-TBDI1-29/C1S4-31
P1-TBDI1-31	C1/C1S4	C1S4-33/P1-TBDI1-31	P1-TBDI1-31/C1S4-33
P1-TBDI2-1	C1/C1S5	C1S5-1/P1-TBDI2-1	P1-TBDI2-1/C1S5-1
P1-TBDI2-3	C1/C1S5	C1S5-3/P1-TBDI2-3	P1-TBDI2-3/C1S5-3
P1-TBDI2-5	C1/C1S5	C1S5-5/P1-TBDI2-5	P1-TBDI2-5/C1S5-5
P1-TBDI2-7	C1/C1S5	C1S5-7/P1-TBDI2-7	P1-TBDI2-7/C1S5-7
P1-TBDI2-9	C1/C1S5	C1S5-9/P1-TBDI2-9	P1-TBDI2-9/C1S5-9
P1-TBDI2-11	C1/C1S5	C1S5-11/P1-TBDI2-11	P1-TBDI2-11/C1S5-11
P1-TBDI2-12	C1/C1S5	C1S5-12/P1-TBDI2-12	P1-TBDI2-12/C1S5-12
P1-TBDI2-14	C1/C1S5	C1S5-14/P1-TBDI2-14	P1-TBDI2-14/C1S5-14
P1-TBDI2-16	C1/C1S5	C1S5-16/P1-TBDI2-16	P1-TBDI2-16/C1S5-16
P1-TBDI2-18	C1/C1S5	C1S5-20/P1-TBDI2-18	P1-TBDI2-18/C1S5-20
P1-TBDI2-21	C1/C1S5	C1S5-23/P1-TBDI2-21	P1-TBDI2-21/C1S5-23
P1-TBDI2-23	C1/C1S5	C1S5-25/P1-TBDI2-23	P1-TBDI2-23/C1S5-25
P1-TBDI2-25	C1/C1S5	C1S5-27/P1-TBDI2-25	P1-TBDI2-25/C1S5-27
P1-TBDI2-27	C1/C1S5	C1S5-29/P1-TBDI2-27	P1-TBDI2-27/C1S5-29
P1-TBDI2-29	C1/C1S5	C1S5-31/P1-TBDI2-29	P1-TBDI2-29/C1S5-31
P1-TBDI2-31	C1/C1S5	C1S5-33/P1-TBDI2-31	P1-TBDI2-31/C1S5-33
P1-TBDI3-1	C1/C1S6	C1S6-1/P1-TBDI3-1	P1-TBDI3-1/C1S6-1
P2-TBDI1-7	C2/C2S1	C2S1-7/P2-TBDI1-7	P2-TBDI1-7/C2S1-7
P2-TBDI1-9	C2/C2S1	C2S1-9/P2-TBDI1-9	P2-TBDI1-9/C2S1-9
P2-TBDI1-11	C2/C2S1	C2S1-11/P2-TBDI1-11	P2-TBDI1-11/C2S1-11
P2-TBDI1-13	C2/C2S1	C2S1-13/P2-TBDI1-13	P2-TBDI1-13/C2S1-13
P2-TBDI1-15	C2/C2S1	C2S1-15/P2-TBDI1-15	P2-TBDI1-15/C2S1-15
P2-TBDI1-17	C2/C2S1	C2S1-19/P2-TBDI1-17	P2-TBDI1-17/C2S1-19
P2-TBDI1-19	C2/C2S1	C2S1-21/P2-TBDI1-19	P2-TBDI1-19/C2S1-21

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P1-TBDI3-17	C1/C1S6	C1S6-19/P1-TBDI3-17	P1-TBDI3-17/C1S6-19
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P3-TBAI2-1(B)	C3/C3S7	C3S7-1/P3-TBAI2-1(B)	P3-TBAI2-1(B)/C3S7-1
P3-TBAI3-4(B)	C3/C3S8	C3S8-7/P3-TBAI3-4(B)	P3-TBAI3-4(B)/C3S8-7
P1-TBDI4-18	C1/C1S7	C1S7-20/P1-TBDI4-18	P1-TBDI4-18/C1S7-20
P2-TBDI3-9	C2/C2S3	C2S3-9/P2-TBDI3-9	P2-TBDI3-9/C2S3-9
P1-TBDI3-20	C1/C1S6	C1S6-22/P1-TBDI3-20	P1-TBDI3-20/C1S6-22
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P1-TBDI3-22	C1/C1S6	C1S6-24/P1-TBDI3-22	P1-TBDI3-22/C1S6-24
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-	-	-	-
P1-TBDI3-24	C1/C1S6	C1S6-26/P1-TBDI3-24	P1-TBDI3-24/C1S6-26
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-	-	-	-
P3-TBDI3-9	C3/C3S3	C3S3-9/P3-TBDI3-9	P3-TBDI3-9/C3S3-9
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P3-TBAI3-10(B)	C3/C3S8	C3S8-21/P3-TBAI3-10(B)	P3-TBAI3-10(B)/C3S8-21
P2-TBDI3-25	C2/C2S3	C2S3-27/P2-TBDI3-25	P2-TBDI3-25/C2S3-27
P2-TBDI3-29	C2/C2S3	C2S3-31/P2-TBDI3-29	P2-TBDI3-29/C2S3-31
P2-TBDI4-1	C2/C2S4	C2S4-1/P2-TBDI4-1	P2-TBDI4-1/C2S4-1
P2-TBDI4-5	C2/C2S4	C2S4-5/P2-TBDI4-5	P2-TBDI4-5/C2S4-5
P2-TBDI4-9	C2/C2S4	C2S4-9/P2-TBDI4-9	P2-TBDI4-9/C2S4-9
P2-TBDI4-11	C2/C2S4	C2S4-11/P2-TBDI4-11	P2-TBDI4-11/C2S4-11
P2-TBDI4-13	C2/C2S4	C2S4-13/P2-TBDI4-13	P2-TBDI4-13/C2S4-13

P2-TBDI4-15	C2/C2S4	C2S4-15/P2-TBDI4-15	P2-TBDI4-15/C2S4-15
P2-TBDI4-19	C2/C2S4	C2S4-21/P2-TBDI4-19	P2-TBDI4-19/C2S4-21
P2-TBDI4-21	C2/C2S4	C2S4-23/P2-TBDI4-21	P2-TBDI4-21/C2S4-23
P2-TBDI4-23	C2/C2S4	C2S4-25/P2-TBDI4-23	P2-TBDI4-23/C2S4-25
P2-TBDI4-25	C2/C2S4	C2S4-27/P2-TBDI4-25	P2-TBDI4-25/C2S4-27
P2-TBDI4-27	C2/C2S4	C2S4-29/P2-TBDI4-27	P2-TBDI4-27/C2S4-29
P2-TBDI4-29	C2/C2S4	C2S4-31/P2-TBDI4-29	P2-TBDI4-29/C2S4-31
P2-TBDI4-31	C2/C2S4	C2S4-33/P2-TBDI4-31	P2-TBDI4-31/C2S4-33
P2-TBDI5-1	C2/C2S5	C2S5-1/P2-TBDI5-1	P2-TBDI5-1/C2S5-1
P2-TBDI5-5	C2/C2S5	C2S5-5/P2-TBDI5-5	P2-TBDI5-5/C2S5-5
P2-TBDI5-7	C2/C2S5	C2S5-7/P2-TBDI5-7	P2-TBDI5-7/C2S5-7
P2-TBDI5-9	C2/C2S5	C2S5-9/P2-TBDI5-9	P2-TBDI5-9/C2S5-9
P2-TBDI5-11	C2/C2S5	C2S5-11/P2-TBDI5-11	P2-TBDI5-11/C2S5-11
P2-TBDI5-13	C2/C2S5	C2S5-13/P2-TBDI5-13	P2-TBDI5-13/C2S5-13
P2-TBDI5-15	C2/C2S5	C2S5-15/P2-TBDI5-15	P2-TBDI5-15/C2S5-15
P2-TBDI5-17	C2/C2S5	C2S5-19/P2-TBDI5-17	P2-TBDI5-17/C2S5-19
P2-TBDI5-19	C2/C2S5	C2S5-21/P2-TBDI5-19	P2-TBDI5-19/C2S5-21
P2-TBDI3-13	C2/C2S3	C2S3-13/P2-TBDI3-13	P2-TBDI3-13/C2S3-13
P2-TBDI3-15	C2/C2S3	C2S3-15/P2-TBDI3-15	P2-TBDI3-15/C2S3-15
P2-TBDI3-19	C2/C2S3	C2S3-21/P2-TBDI3-19	P2-TBDI3-19/C2S3-21
P2-TBDI3-23	C2/C2S3	C2S3-25/P2-TBDI3-23	P2-TBDI3-23/C2S3-25
P2-TBDI3-27	C2/C2S3	C2S3-29/P2-TBDI3-27	P2-TBDI3-27/C2S3-29
P2-TBDI3-31	C2/C2S3	C2S3-33/P2-TBDI3-31	P2-TBDI3-31/C2S3-33
P2-TBDI4-3	C2/C2S4	C2S4-3/P2-TBDI4-3	P2-TBDI4-3/C2S4-3
P2-TBDI4-7	C2/C2S4	C2S4-7/P2-TBDI4-7	P2-TBDI4-7/C2S4-7
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P2-TBDI2-22	C2/C2S2	C2S2-24/P2-TBDI2-22	P2-TBDI2-22/C2S2-24
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P4-TBDI3-7	C4/C4S3	C4S3-7/P4-TBDI3-7	P4-TBDI3-7/C4S3-7
P4-TBDI3-8	C4/C4S3	C4S3-8/P4-TBDI3-8	P4-TBDI3-8/C4S3-8
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P4-TBDI2-7	C4/C4S2	C4S2-7/P4-TBDI2-7	P4-TBDI2-7/C4S2-7
P4-TBDI2-8	C4/C4S2	C4S2-8/P4-TBDI2-8	P4-TBDI2-8/C4S2-8
P4-TBDI3-9	C4/C4S3	C4S3-9/P4-TBDI3-9	P4-TBDI3-9/C4S3-9
P4-TBDI3-10	C4/C4S3	C4S3-10/P4-TBDI3-10	P4-TBDI3-10/C4S3-10
P4-TBDI3-11	C4/C4S3	C4S3-11/P4-TBDI3-11	P4-TBDI3-11/C4S3-11
P4-TBDI2-1	C4/C4S2	C4S2-1/P4-TBDI2-1	P4-TBDI2-1/C4S2-1
P4-TBDI2-2	C4/C4S2	C4S2-2/P4-TBDI2-2	P4-TBDI2-2/C4S2-2
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P4-TBDI1-1	C4/C4S1	C4S1-1/P4-TBDI1-1	P4-TBDI1-1/C4S1-1
P4-TBDI1-2	C4/C4S1	C4S1-2/P4-TBDI1-2	P4-TBDI1-2/C4S1-2
P4-TBDI1-3	C4/C4S1	C4S1-3/P4-TBDI1-3	P4-TBDI1-3/C4S1-3
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P3-TBDI2-3	C3/C3S2	C3S2-3/P3-TBDI2-3	P3-TBDI2-3/C3S2-3
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P2-TBRL1-17	C2/C2S6	C2S6-19/P2-TBRL1-17	P2-TBRL1-17/C2S6-19
P2-TBDI2-12	C2/C2S2	C2S2-12/P2-TBDI2-12	P2-TBDI2-12/C2S2-12
P2-TBDI2-13	C2/C2S2	C2S2-13/P2-TBDI2-13	P2-TBDI2-13/C2S2-13
P1-TBAI2-1(B)	C1/C1S11	C1S11-1/P1-TBAI2-1(B)	P1-TBAI2-1(B)/C1S11-1
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P2-TBDI2-14	C2/C2S2	C2S2-14/P2-TBDI2-14	P2-TBDI2-14/C2S2-14
P2-TBDI2-15	C2/C2S2	C2S2-15/P2-TBDI2-15	P2-TBDI2-15/C2S2-15
P2-TBDI2-6	C2/C2S2	C2S2-6/P2-TBDI2-6	P2-TBDI2-6/C2S2-6
P2-TBDI2-9	C2/C2S2	C2S2-9/P2-TBDI2-9	P2-TBDI2-9/C2S2-9
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P2-TBDI2-16	C2/C2S2	C2S2-16/P2-TBDI2-16	P2-TBDI2-16/C2S2-16
P2-TBDI2-17	C2/C2S2	C2S2-19/P2-TBDI2-17	P2-TBDI2-17/C2S2-19
P2-TBDI2-7	C2/C2S2	C2S2-7/P2-TBDI2-7	P2-TBDI2-7/C2S2-7
P2-TBDI2-10	C2/C2S2	C2S2-10/P2-TBDI2-10	P2-TBDI2-10/C2S2-10

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P1-TBDI3-17	C1/C1S6	C1S6-19/P1-TBDI3-17	P1-TBDI3-17/C1S6-19
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P1-TBAI1-11(B)	C1/C1S10	C1S10-23/P1-TBAI1-11(B)	P1-TBAI1-11(B)/C1S10-23
P1-TBDI3-18	C1/C1S6	C1S6-20/P1-TBDI3-18	P1-TBDI3-18/C1S6-20
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P1-TBDI3-26	C1/C1S6	C1S6-28/P1-TBDI3-26	P1-TBDI3-26/C1S6-28
P1-TBDI4-16	C1/C1S7	C1S7-16/P1-TBDI4-16	P1-TBDI4-16/C1S7-16
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-	-	-	-
-	-	-	-
P1-TBDI4-22	C1/C1S7	C1S7-24/P1-TBDI4-22	P1-TBDI4-22/C1S7-24
P1-TBDI4-23	C1/C1S7	C1S7-25/P1-TBDI4-23	P1-TBDI4-23/C1S7-25
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P5-TBDI1-8	C5/C5S1	C5S1-8/P5-TBDI1-8	P5-TBDI1-8/C5S1-8
P5-TBDI1-9	C5/C5S1	C5S1-9/P5-TBDI1-9	P5-TBDI1-9/C5S1-9
P5-TBDI1-10	C5/C5S1	C5S1-10/P5-TBDI1-10	P5-TBDI1-10/C5S1-10
P5-TBDI1-11	C5/C5S1	C5S1-11/P5-TBDI1-11	P5-TBDI1-11/C5S1-11
P5-TBDI1-12	C5/C5S1	C5S1-12/P5-TBDI1-12	P5-TBDI1-12/C5S1-12
P5-TBDI1-13	C5/C5S1	C5S1-13/P5-TBDI1-13	P5-TBDI1-13/C5S1-13
P5-TBDI1-14	C5/C5S1	C5S1-14/P5-TBDI1-14	P5-TBDI1-14/C5S1-14
P5-TBDI1-27	C5/C5S1	C5S1-29/P5-TBDI1-27	P5-TBDI1-27/C5S1-29
P5-TBDI1-29	C5/C5S1	C5S1-31/P5-TBDI1-29	P5-TBDI1-29/C5S1-31
P5-TBDI1-31	C5/C5S1	C5S1-33/P5-TBDI1-31	P5-TBDI1-31/C5S1-33
P5-TBDI2-1	C5/C5S2	C5S2-1/P5-TBDI2-1	P5-TBDI2-1/C5S2-1
P5-TBDI2-3	C5/C5S2	C5S2-3/P5-TBDI2-3	P5-TBDI2-3/C5S2-3
P5-TBDI2-5	C5/C5S2	C5S2-5/P5-TBDI2-5	P5-TBDI2-5/C5S2-5
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
P5-TBDI1-2	C5/C5S1	C5S1-2/P5-TBDI1-2	P5-TBDI1-2/C5S1-2
P5-TBAI1-6(B)	C5/C5S4	C5S4-13/P5-TBAI1-6(B)	P5-TBAI1-6(B)/C5S4-13
P5-TBAI1-7(B)	C5/C5S4	C5S4-15/P5-TBAI1-7(B)	P5-TBAI1-7(B)/C5S4-15
P5-TBDI2-7	C5/C5S2	C5S2-7/P5-TBDI2-7	P5-TBDI2-7/C5S2-7
-	-	-	-
P5-TBDI1-15	C5/C5S1	C5S1-15/P5-TBDI1-15	P5-TBDI1-15/C5S1-15
P5-TBDI1-16	C5/C5S1	C5S1-16/P5-TBDI1-16	P5-TBDI1-16/C5S1-16
P5-TBDI1-17	C5/C5S1	C5S1-19/P5-TBDI1-17	P5-TBDI1-17/C5S1-19
P5-TBDI1-18	C5/C5S1	C5S1-20/P5-TBDI1-18	P5-TBDI1-18/C5S1-20
P5-TBDI1-19	C5/C5S1	C5S1-21/P5-TBDI1-19	P5-TBDI1-19/C5S1-21
-	-	-	-
-	-	-	-
-	-	-	-
P5-TBDI1-21	C5/C5S1	C5S1-23/P5-TBDI1-21	P5-TBDI1-21/C5S1-23
P5-TBDI1-22	C5/C5S1	C5S1-24/P5-TBDI1-22	P5-TBDI1-22/C5S1-24
P5-TBDI1-23	C5/C5S1	C5S1-25/P5-TBDI1-23	P5-TBDI1-23/C5S1-25

P5-TBDI1-24	C5/C5S1	C5S1-26/P5-TBDI1-24	P5-TBDI1-24/C5S1-26
P5-TBDI1-25	C5/C5S1	C5S1-27/P5-TBDI1-25	P5-TBDI1-25/C5S1-27
-	-	-	-
-	-	-	-

PLC Tag Name	Cold Continuity (<2Ω)	Cold Megger (>20MΩ)	Hot Power (24VDC)	Calibration 0% (4mA)
AT-60 241_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
DPT-60 201_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
DPT-61 301_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
DPT-61 302_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
DPT-61 310_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
DPT-61 312_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
DPT-60 202_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
DPT-60 203_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
FSL-40 201_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FSL-40 202_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FS-40 204_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FS-40 205_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-40 201_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-40 202_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-40 221_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-40 222_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 221_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 201_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 210_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 211_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 261FA_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 261FB_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 261FC_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 261FD_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 262FE_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 262FF_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 262FG_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 262FH_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 263FJ_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 263FK_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA

FT-60 263FL_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 263FM_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 264FN_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 264FP_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 264FQ_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 264FR_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-61 307_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 203_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 202_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 204_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-936 201_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-936 202_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
HV-40 203_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
HV-40 201_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
HV-40 204_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
HV-40 202_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
HV-40 206_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA

HV-40 205_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
HV-40 207_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
HV-40 208_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LSH-60 230_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LSH-60 231_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LSH-60 240_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LSH-60 250_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LSH-60 251_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LSL-60 201_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LSH-61 301_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LSL-61 302_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LS-61 303_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LSH-61 304_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LSH-61 305_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LSH-61 306_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LSH-61 307_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LSH-61 308_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LSH-61 312_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LSL-814 201_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LT-40 201_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
LT-40 202_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
LT-40 203_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
LT-61 301_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
pHT-40 201_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
pHT-40 202_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PSL-60 220_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
PSL-61 301_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
PSL-61 302_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
PSL-61 305_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
PSL-61 306_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
PSL-61 311_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-40 201_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-40 202_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-40 222_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-60 201_AI_1	[] Pass	____ MΩ	[] 24V	____ mA

PT-60 202_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-60 203_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-60 210_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-60 215_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-60 216_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-60 220_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-60 221_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-60 230_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-60 231_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-60 240_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-60 241_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-60 204_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-61 301_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-61 302_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-61 303_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-61 304_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-61 305_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-61 309_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PV-60 201_AO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-40 201_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-40 202_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-40 203_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-40 204_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-60 220_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-60 230_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-60 231_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-60 240A_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-60 240B_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-60 240C_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-61 301_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-61 302_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-61 306_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-61 303_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-61 305_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-61 308_DO_1	[] Pass	____ MΩ	[] 24V	____ mA

SV-61 309_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-61 310_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
SV-61 311_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
TSH-60 230_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
TSH-60 240_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-40 221_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-40 222_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 201_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 210_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 210A_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 210B_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 211_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 214_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 215A_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 215B_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 216_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 220_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 221_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 222_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 223_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 224_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 225_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 226_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 227_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 230_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 240_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 241A_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 241B_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 245_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-60 202_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-61 301_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TT-61 302_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
TCV-40 222_AO_1	[] Pass	____ MΩ	[] 24V	____ mA
TV-60 214_AO_1	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA

VT-60 210A_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
VT-60 210B_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
VT-60 241A_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
VT-60 241B_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
XT-60 201_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
XA-60 201_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
XT-92 201_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
XA-92 201_DO_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 201A_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 201B_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 201C_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 201D_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 201E_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 201F_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 201G_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 201H_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 202A_DI_2	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 202B_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 202C_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 202D_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 202E_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 202F_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 202G_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 202H_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 203B_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 203C_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 203D_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 203E_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 203F_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 204B_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 204C_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 205A_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 205B_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 205C_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-40 205D_DI_1	[] Pass	____ MΩ	[] 24V	____ mA

[illegible]

-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
ES-60 202_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
FT-60 207_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
PT-60 207_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
XV-60 240D_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
SV-60 207_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
SV-60 201_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
SV-60 202_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
SV-60 203_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
SV-61 304_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
TT-60 207_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
XV-60 261FA_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
XV-60 261FB_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
XV-60 261FC_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
XV-60 261FD_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
XV-60 261SA_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
XV-60 261SB_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
XV-60 261SC_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA

XV-60 261SD_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 262FE_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 262FF_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 262FG_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 262FH_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 262SE_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 262SF_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 262SG_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 262SH_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 263FJ_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 263FK_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 263FL_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 263FM_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 263SJ_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 263SK_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 263SL_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 263SM_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 264FN_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 264FP_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 264FQ_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 264FR_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 264SN_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 264SP_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 264SQ_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 264SR_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA
XV-60 241_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA
PSE-61 300_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
PSE-61 301_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA

[illegible]

[illegible]

-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
XS-60 240B_DO_1	[] Pass	_____ MΩ	[] 24V	_____ mA
XA-60 240A_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
XA-60 240B_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
DPI-60 240_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
XA-60 241A_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
XA-60 241B_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
TSH-60 241A_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
TSL-60 241B_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
XA-60 242A_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
XA-60 242B_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
TSH-60 242A_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
TSL-60 242B_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA

[illegible]

-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
ES-60 202_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
PT-60 202_AI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
FS-60 205_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
VS-60 205_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
XV-60 205A_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA
YA-60 205_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
YI-60 205_DI_1	[] Pass	_____ MΩ	[] 24V	_____ mA
-	[] Pass	_____ MΩ	[] 24V	_____ mA

[illegible]

HV-20 204_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
HV-20 205_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
HV-20 206_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
HV-20 207_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
HV-20 208_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
HV-20 209_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
HV-20 210_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-20 201_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-20 202_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-20 203_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-20 211_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-20 212_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-20 213_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA
FT-913 201_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
LT-913 201_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
PT-913 201_AI_1	[] Pass	____ MΩ	[] 24V	____ mA
XV-913 201_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA
XA-80 305A_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XA-80 305B_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XA-80 305C_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XA-80 305D_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XA-80 305E_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA
-	[] Pass	____ MΩ	[] 24V	____ mA
XA-80 306A_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XA-80 306B_DI_1	[] Pass	____ MΩ	[] 24V	____ mA
XA-80 306C_DI_1	[] Pass	____ MΩ	[] 24V	____ mA

XA-80 306D_DI_1	[] Pass	---- MΩ	[] 24V	---- mA
XA-80 306E_DI_1	[] Pass	---- MΩ	[] 24V	---- mA
-	[] Pass	---- MΩ	[] 24V	---- mA
-	[] Pass	---- MΩ	[] 24V	---- mA

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

____ mA	____ mA	[] Pass	READY FOR TEST	Pending Sign-off
____ mA	____ mA	[] Pass	READY FOR TEST	Pending Sign-off
____ mA	____ mA	[] Pass	RESERVE ALLOC	Pending Sign-off
____ mA	____ mA	[] Pass	RESERVE ALLOC	Pending Sign-off

Item #	Activity / Inspection Description	Reference Standard / Spec
1	Receiving Inspection of Field Instruments & Transmitters	Project Instrument Datasheets
2	Cable Tray, Conduit & Field Gland Integrity Inspection	IEC 60079-14 / NEC 500
3	Field Instrument Mechanical Mounting & Impulse Piping	P&ID / Hook-up Drawings
4	Cable Continuity & Insulation Resistance (Megger Test)	IEC 62381 Clause 5.2
5	Shield Single-Point Earthing & Barrier Ground Bar	ISA-RP60.8 / IEC 60079-11
6	Hot Loop 24VDC Power Verification & Polarity Check	ControlLogix Wiring Diagram
7	Analog Transmitter 5-Point Calibration Simulation (AI)	IEC 62381 Clause 5.3
8	Control Valve Modulation Stroke & Fail-Safe Test (AO)	ISA-75.05 / IEC 60534
9	Discrete Limit Switch & Solenoid Valve Function Check	P&ID & Logic Diagram
10	Safety Interlock Trip & Emergency Shutdown (ESD) Test	Process Cause & Effect Matrix
11	SCADA Graphic HMI Representation & Alarm Logging	SCADA Architecture Spec
12	Punchlist Clearance (Category A Zero-Defect Handover)	Contract Commissioning Spec

ENGINEERING | INSPECTION & TEST PLAN (ITP) — INSTRUMENTATION & CONTROL SYSTEMS

Quality Assurance & Control Verification Matrix (H = Hold Point, W = Witness Point, R = Review of Records)

Acceptance Criteria	Verifying Document	Contractor (Kalasin)
Damage-free, correct model, serial no., tag plate	<i>Material Receiving Report (MRR)</i>	H (Perform)
Proper earthing, fire stops, Ex d/e rated glands	<i>Cable Installation Record</i>	H (Perform)
Vibration-free, correct slopes, 5-valve manifolds tight	<i>Mechanical Completion Cert</i>	H (Perform)
Core-Core & Core-Earth > 20 MΩ @ 500VDC	<i>Cold Loop Check Sheet</i>	H (Perform)
Single-point earth at CA1 only. IS ground < 1.0 Ω	<i>Earthing Test Certificate</i>	H (Perform)
Voltage 24VDC ±10%. Correct polarity without shorts	<i>Power-Up Inspection Log</i>	H (Perform)
Accuracy ≤ ±0.5% span at 0, 25, 50, 75, 100%	<i>Loop Calibration Record</i>	H (Perform)
Stroke error ≤ 1.0%. Fail-safe action < 2 sec	<i>Valve Stroke Test Sheet</i>	H (Perform)
100% position switch fidelity & travel time < 3 sec	<i>Discrete Check Sheet</i>	H (Perform)
100% trip execution, alarm banner flashing, horn	<i>Safety Trip Protocol</i>	H (Perform)
Real-time updates, engineering units, alarm logs	<i>HMI Verification Log</i>	H (Perform)
Zero Category A items. Cat B clearance schedule agreed	<i>Punchlist Clearance Form</i>	H (Perform)

S

Client (Ingredion)	Third Party / TUV	Milestone Stage
W (Witness)	R (Review)	Pre-Installation
W (Witness)	R (Review)	Pre-Commissioning
W (Witness)	R (Review)	Pre-Commissioning
W (Witness)	R (Review)	Loop Test Phase 1
H (Hold)	W (Witness)	Loop Test Phase 1
R (Review)	R (Review)	Loop Test Phase 2
W (Witness)	R (Review)	Loop Test Phase 3
H (Hold)	R (Review)	Loop Test Phase 4
W (Witness)	R (Review)	Loop Test Phase 4
H (Hold)	H (Hold)	Loop Test Phase 5
W (Witness)	R (Review)	Loop Test Phase 5
H (Hold)	R (Review)	Final Handover

KALASIN ENGINEERING	
PROJECT: SPRINT 18K THERMAL OIL	

Tag Number:	FT-40 201	P&ID Number:	FT-40 201
Signal Type:	4 - 20 mA HART	Calibrated Range:	0.0 - 15,000.0 kg/hr
Junction Box:	JB-401 (Floor 2F)	Marshaling TB:	JB-401 TB
PLC Tag Name:	FT-40 201_AI_1	Manufacturer:	Endress+Hauser

TEST EQUIPMENT & CALIBRATOR TRACEABILITY		
Equipment Type	Manufacturer / Model	Serial Number
Documenting Process Calibrator	Fluke 754	FLK-754-98421

5-POINT ANALOG LOOP SIMULATION & CALIBRATION RESULTS		
Test Point (%)	Input Signal (mA)	Expected Value (EU)
0.0 % (Zero)	4.00 mA	0.0 kg/hr
25.0 % (Quarter)	8.00 mA	3,750.0 kg/hr
50.0 % (Mid-scale)	12.00 mA	7,500.0 kg/hr
75.0 % (Three-Quarter)	16.00 mA	11,250.0 kg/hr
100.0 % (Full Span)	20.00 mA	15,000.0 kg/hr

TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF		
Sign-off Role	Name & Title	Company / Organization
Tested by (Contractor):	Lead Instrument Engineer	Kalasin Engineering Co., Ltd.
Witnessed by (Client QC):	Senior Electrical & Inst QA/QC	Ingredion (Thailand) Co., Ltd.
Approved by (Commissioning PM):	Project Commissioning Manager	Kalasin / Ingredion Joint Team

NEERING | INSTRUMENT LOOP CHECK CERTIFICATE

PA SPRAY DRYER / JET COOKER | CLIENT: INGREDION (THAILAND) CO., LTD.

325-01-402-PD-02	Service Desc: Slurry Infeed Mass Flow Transmitter
0 - 15,000 kg/hr	Eng. Units: kg/hr
P1-TBDI1-19	PLC Channel: Chassis C1 / Slot 4 / Pin 21
Endress+Hauser	Model Series: Promass F 300 / 83F Coriolis

Calibration Expiry Date	Accuracy	Cal Certificate #
31-DEC-2026	±0.01% Reading	NIST-2026-98102

Field Meas (mA)	SCADA Reading (EU)	PLC Raw Counts	Error (% Span)
4.001 mA	0.0 kg/hr	0 counts	+0.01 %
8.002 mA	3,751.2 kg/hr	8192 counts	+0.02 %
11.998 mA	7,498.5 kg/hr	16384 counts	-0.01 %
16.003 mA	11,252.1 kg/hr	24576 counts	+0.03 %
19.997 mA	14,996.8 kg/hr	32767 counts	-0.02 %

Signature	Date Signed	Overall Verdict	Punchlist Ref
_____	DD-MM-2026	LOOP CHECK PASSED	None (Clean)
_____	DD-MM-2026	ACCEPTED & APPROVED	None
_____	DD-MM-2026	FINAL AUTHORIZED	None

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Result (Pass/Fail)
PASS
PASS
PASS
PASS
PASS

Remarks
Green Loop Tag Affixed
Ready for Wet Commissioning
System Cleared for Operation

1. CAD SOURCE DRAWINGS & DXF CONVERSION MASTER AUDIT

Drawing Filename	File Format	File Size
LOOP DIAGRAM WRING.dwg	AutoCAD DWG (v2018)	2.05 MB
LOOP DIAGRAM WRING.dxf	AutoCAD DXF (ASCII)	28.37 MB
PANEL WIRING DIAGRAM.dwg	AutoCAD DWG (v2018)	4.53 MB
PANEL WIRING DIAGRAM.dxf	AutoCAD DXF (ASCII)	100.82 MB

2. 3-ZONE INSTRUMENT WIRING ARCHITECTURE & TERMINAL DESIGNATIONS

Physical Zone	CAD Drawing Block / Column	Terminal Prefix
Zone 1: FIELD	FIELD column (Transmitter, Valve, Switch)	Terminals 1(+), 2(-), PE
Zone 2: JUNCTION BOX	JUNCTION BOX column (JB-401..618, IS-JB)	TBAI, TBAO, TBDI, TBDO, TBCI
Zone 3: MARSHALLING	CONTROL ROOM column (Cabinet CA1 / RIO-200)	TBDAI, TBDAO, TBDDI, TBDDO
Zone 3: PLC I/O RACK	1756 ControlLogix / 1794 FLEX I/O	RTB Terminal (Chassis C1..C5)

3. MASTER CAD TYPICAL LOOP WIRING CONFIGURATIONS & COMMISSIONING PROCEDURES

CAD Typical ID	Instrument Loop Category	Sample Project Tags
TYP-AI-01	2-Wire Loop Powered Analog Transmitter	PT-10 080, LT-10 081, TT-10 085, PDT-60 101
TYP-AI-02	4-Wire Active Powered Analog Transmitter	FT-102 900, FT-102 901, FT-106 003, DT-106 004
TYP-AO-01	4-20mA SMART Control Valve Positioner	FCV-102 900-1, FCV-102 901-1, FCV-106 003-1
TYP-DI-01	Discrete Status / Limit / Level Switch	LSH-60 101, LSL-60 102, ZSO-102 900, ZSC-102 900
TYP-DO-01	Solenoid Valve / Interposing Relay Drive	XV-102 900, SOV-102 901, SV-60 101, PMP-40 101
TYP-IS-01	Intrinsically Safe Ex Loop (Hazardous Area)	Loops landed in IS-JB-603, 608, 612, 618

4. MARSHALLING CABINET (CA1) INTERNAL ARRANGEMENT & 24VDC POWER SYSTEM

Subsystem Component	Specification / Model in DWG	Location in CA1
Primary Power Supplies (PSU1 & PSU2)	Phoenix Contact Quint Power 24VDC / 40A	Cabinet Base DIN Rail (Left & Right)
Branch Circuit Fuses	Weidmüller WSI 6-LD / WDU 2.5 with 0.5A fast fuse	Marshalling Terminal Rails TBX204..TBX207
Dual Earthing Busbars	PE (Protective Earth) & IE (Instrument Earth)	Cabinet Bottom Isolated Copper Bars (PE=Direct, IE=Isolated)

Internal Wire Ducting	Slotted PVC Trunking 40x100mm & 70x100mm	Vertical & Horizontal Cabinet Wireways
PLC Remote I/O Adapters	1794-AENTR / 1756-EN2TR Dual Port EtherNet/IP	Chassis Mounting Plates

KALASIN ENGINEERING | CAD INSTRUMENT LOOP DIAGRAM & PANEL WIRING S

Application of LOOP DIAGRAM WRING.dwg & PANEL WIRING DIAGRAM.dwg (DWG & DXF) to Project Spr

Modelspace Entities	Text Annotations	Block Definitions
48,670 entities	18,638 texts	1,138 blocks
48,670 entities	18,638 texts	1,138 blocks
22,994 entities	9,462 texts	1,107 blocks
22,994 entities	9,462 texts	1,107 blocks

Terminal Model / Spec	Wire Color Standard	Core / Pair Tagging
M20 / 0.5-inch NPT Ex-d/Ex-e Gland	White (+), Black (-)	Branch 1Px0.75mm ² or 1Px1.5mm ²
Weidmüller WDU 2.5 / Phoenix UK	White (+), Black (-)	PR 1..PR 24 (Pair numbers marked)
WSI 6-LD (0.5A Fuse) / Disconnect	White (+), Black (-), Red/Blue	Multi-pair 24Px0.75mm ² Re-2Y(st)-Yv
1756-TBCH / 1794-TB3 Removable TB	Grey (DC+), Blue (DC-), White (Sig)	Pre-formed wiring harness to CA1

Terminal Strip	Signal Standard	Power Source
TBAI.xx	4-20mA DC (2-Wire)	24VDC from CA1 Marshalling
TBAI.xx	4-20mA DC (Active Output)	220VAC / 24VDC Aux Power
TBAO.xx	4-20mA Command + Travel FB	4-20mA loop powered positioner
TBDI.xx	24VDC Interrogation Contact	24VDC Wetting Voltage from CA1
TBDO.xx	24VDC Switched Output	24VDC from 1756-OB32 / Omron Relay
TB-IS.xx (Blue)	Ex-ia / Ex-ib Certified Loop	Galvanic Isolator (MTL / P+F) in CA1

Redundancy / Protection Mechanism	Testing & Verification Requirement
N+1 Active Redundancy via Diode Decoupling Module	Simulate AC supply failure on PSU1; verify seamless 24VDC transfer without PLC dip
Individual channel fuse isolation with blown-fuse LED	Measure loop voltage drop across fuse holder (<0.1VDC); verify fuse LED indicator
Single-Point Grounding architecture prevents ground loop hum	Measure resistance between IE bar and clean earth pit (<1.0Ω); check IE-to-PE isolation (>10MΩ)

Physical segregation between 24VDC signal and 220VAC power	<i>Inspect 50mm separation between IS blue ducts and non-IS grey ducts; 50% fill factor</i>
Device Level Ring (DLR) fault-tolerant network topology	<i>Perform network ring break test; verify zero packet loss to central ControlLogix processor</i>

SPECIFICATION

Print 18K TPA

Conversion Tool	Engineering Application Scope
Native Client Source	Field-to-Marshalling 3-Zone Loop Typicals (AI, AO, DI, DO, IS)
dwg2dxf (GNU LibreDWG)	Fully Parsable Open CAD Format / ezdxf Programmatic Automation
Native Client Source	Marshalling CA1 & RIO-200 Internal Wiring, Fuses, Diode Redundancy
dwg2dxf (GNU LibreDWG)	Full High-Resolution Vector Export for Electrical CAD Integration

Shield Treatment Philosophy	Inspection Checkpoint
CUTBACK & TAPE (Floating, isolated from instrument housing)	Verify cable gland IP66/67, seal, O-ring, and floating shield
SH Feedthrough (Isolated rail, NOT bonded to JB enclosure frame)	Check terminal torque 0.6Nm, wire ferrules, trunk gland earthing
IE Instrument Earth (Single-Point Star bonded to Clean Earth Pit)	Verify fuse rating (0.5A fast), disconnect knife closed, IE <1.0Ω
Internal chassis backplane isolated ground	Verify card LED active, channel OK, tag scaled in ControlLogix

Commissioning Cold Check	Commissioning Hot / Calibration Check
Megger core-to-earth >20MΩ @ 500VDC, continuity <2.0Ω	Inject 4, 8, 12, 16, 20 mA via Fluke 789 at field terminals; verify ControlLogix tag scaling ±0.2%
Megger power and signal cores separately >20MΩ	Power up instrument; measure 24VDC/220VAC; simulate 0-100% flow from local transmitter menu; check SCADA
Loop resistance check (typ. 250-450Ω), air supply 5.5 bar	Force 4, 8, 12, 16, 20 mA from ControlLogix; verify 0, 25, 50, 75, 100% physical stroke and limit switches
Contact open = ∞ MΩ, contact closed = <2.0Ω	Verify 24VDC at field contact; actuate switch manually; verify PLC card LED on/off and HMI alarm latching
Coil resistance check (typ. 30-120Ω), diode polarity check	Command DO bit from ControlLogix; verify interposing relay pull-in, 24VDC at solenoid, pneumatic actuation
Verify blue trunk cable, isolated shield, clearance >50mm	Measure barrier output Voc, Isc; verify entity parameter matching (Ca > C_cable + C_inst, La > L_cable + L_inst)

1. **Introduction**

2. **Methodology**

3. **Results**

4. **Discussion**

5. **Conclusion**